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I still know you were here: Leveraging probe request templates for identifying Wi-Fi devices at scale ☆

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ABSTRACT

MAC address randomization in Wi-Fi Probe Requests (PRs) is supposed to ensure unlinkability for improving user privacy, but PRs still contain enough information to track or re-identify users without relying on the MAC address. This poses a risk to privacy but may also assist law enforcement in identifying devices present at a crime scene. We examine whether it is possible to separate Wi-Fi devices based on observed PRs with randomized MAC addresses using only the content and structure of these PRs. Previous work has predominantly focused on techniques for device counting using feature reduction as a means to manage data set complexity, and has failed to achieve sufficient accuracy for individual device identification. We propose an approach that leverages templates reflecting a PR's structure to identify its influence based on vendor and device, allowing the use of more complex fingerprinting algorithms that utilize the full set of available features. To that end we examine differences between vendors and devices based on observation length, used information elements, and overlapping Fingerprints (FPs) while ignoring MAC addresses. Using PR templating, we construct a knowledge base of FPs and templates from not only existing public data sets but also a new data set published for future research. Our data set tackles critical challenges in labelling and quality in currently available data sets, and we introduce a streamlined and comprehensible crowdsourcing process including automated measurements to enable other researchers to contribute to our data set. We evaluate our device identification approach on the currently available data and demonstrate that, depending on the data set, between 75 % and 85 % of Wi-Fi devices can be uniquely separated within the anonymity group of devices contributing PRs to the respective data set.

1. Introduction

The Wi-Fi network discovery process has long been under scrutiny for compromising user privacy. It involves periodic, active scanning by client devices using Probe Request (PR). These PRs not only announce the device's presence, but also include the MAC address as a unique and permanent device identifier. The situation has improved greatly with the introduction of MAC address randomization, theoretically rendering PRs useless for identification. However, research has since shown that MAC address randomization does not solve all of the existing privacy issues. In part, this is due to the fact that MAC address randomization is not formally standardized, resulting in varying implementations on how and when exactly to randomize MAC addresses. The specific behaviour depends not only on a device's hardware, but also on its operating system and configuration [1]. Older devices may not support MAC randomization at all, while newer devices may force its use or allow the user to enable and disable it. Differences in MAC address randomization behaviour are discussed in more detail in [1]. Ultimately, these privacy measures are supposed to prevent *re-identification* of previously seen devices, which describes the act of using available information to recognize a previously seen device in another situation. Despite MAC address randomization preventing the use of the MAC address for identification, the transmitted PRs may still be used to link ephemeral MAC addresses and thus re-identify individual devices. This is due to the high information content of PRs with many optional and/or custom fields. Previous research [2] refers to this process as *MAC de-randomization*, which is somewhat misleading because it is does

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not actually obtain the permanent hardware MAC address. Instead, PRs are grouped by which device transmitted them. If this grouping is correct, any device can be re-identified by attributing new PRs to its group without requiring knowledge of its permanent MAC address, affecting user privacy to the same extent as true de-randomization The characteristics of a device's PR group can be used to re-identify that device like a Fingerprint (FP), so we refer to this process as *fingerprinting*. The more data can be gathered from an observed device, the sounder is the foundation for later re-identification using its PR FP.

In recent years, the research interest in MAC address fingerprinting and Device Identification (DI) has increased, and more and more data sets have been published. Overall, use cases and application concepts of fingerprinting vary greatly from benign device counting to malicious tracking [2–4]. Wi-Fi technology is of particular interest because of its wide adoption in mobile devices like smartphones, enabling tracking on a large scale. By actively probing for Wi-Fi networks, these devices reveal their presence to keen listeners. Installing such a keen listener could help in guarding private properties against intruders by detecting their Wi-Fi devices and collecting data for later re-identification. This information could give tangible leads to law enforcement or, in turn, compromise the privacy of an innocent user. A successful long-term DI attack would enable an attacker to track a Wi-Fi device anywhere given a comprehensive sensor network to continuously monitor packets transmitted by that device. Such an attack would enable many risks, from locating and apprehending the person carrying the phone, to breaking into this persons home while they are not there. Similarly, criminals could identify Wi-Fi devices carried by the police to stage an early retreat or destroy evidence.

Existing literature acknowledges the importance and usefulness of PRs towards Wi-Fi DI, particularly the usage of Information Elements (IEs) to differentiate sending devices. Still, rarely are attempts made at identifying individual devices, much less at cataloguing vendor or device specific quirks, which would enable a deeper understanding of how PRs are used and how their usage evolves. Due to the complexity of PRs common approaches utilize learning methods like Random Forest to select features and cluster PRs in order to separate devices, with limited success. In this paper we aim to show that structural analysis allows for DI of individual devices in a comprehensive manner.

Our use case precisely aims to detect an unauthorized intruder within a monitored area based on PR DI in Wi-Fi routers. When an unknown mobile device enters, for example, a private residence, the system detects the intrusion and generates an alert. This is enabled by assuming a functioning mechanism that marks a certain PR as *suspicious*, as that PR triggered the alarm, which will not be discussed in this paper.

Other use cases may involve device counting, where efficient and reliable DI mechanisms can be strong assets. Given a gathering of many people, i.e. in a stadium or at an airport, disaster prevention systems may make use of similar systems and provide assistance to prevent overcrowding of certain areas, thus preventing panics. Another application could be tracking ridership and routes taken in public transit networks, allowing the operator to optimize station layouts and schedules according to the passengers' needs.

In this paper, we propose the use of PR templates as an assisting method for the process of DI. Previous approaches have focused in particular on the presence of specific IEs and their parameters, but not exclusively on their structural composition. However, our results show that the structural composition of PRs tells a lot about individual devices but also vendor specific implementations. Previous fingerprinting algorithms suffer from performance constraints, restricting them to using only a limited subset of available features to reduce complexity. Our templating follows a divide-and-conquer approach that splits the large set of PRs into several subsets before applying the fingerprinting algorithm. This effectively reduces the number of PRs that the fingerprinting algorithm must work on, allowing the use of more complex algorithms and the consideration of all available features to improve overall identification accuracy. We are not aware of any other research that emphasized the usage of the PR structure to amplify DI techniques, or examines PR structure on a comparable scale. In addition to investigating how PR templating may assist previous work on DI, we also introduce a novel deterministic template-based DI approach providing comprehensible and explainable results through the use of a knowledge base. A knowledge base that is filled through empirical observations of real world devices and their transmitted PRs, which enables our approach to not only separate devices, but also identify vendor and model in many cases.

We evaluate our DI approach using public PR data sets, but find that existing data sets are insufficient in both quality and scope. To address this, we compile a new data set that contains PRs with randomized MAC addresses and a ground-truth labelling which PR originated from what device. Given the easy reproducibility of our measurement setup, low hardware cost and high degree of automation, we present a crowdsourcing methodology to grow our data set even further.

Finally, this paper answers the following research question: Can a single Wi-Fi device be separated from all other Wi-Fi devices nearby based on the observed PRs? As this type of research strongly relies on available PR data, the secondary research question is: How can we build a PR data set that addresses shortcomings of existing data sets? Our contributions comprise

- an analysis of state of the art PR-based fingerprinting techniques,
- a new data set of PRs captured for devices with MAC address randomization both enabled and disabled,
- automation of PR measurements with Android Debug Bridge (ADB) and work towards a crowd-sourcing process to continue growing our data set,
- an approach utilizing PR templates that can help improve current PR-based fingerprinting techniques.
- a method to build an efficient knowledge base to assist templating-based DI.

2. Background and related work

This section illustrates an example fingerprinting-based detection model, gives some technical background and presents related work. It forms the basis for decisions regarding data selection and experiment design that are presented in later sections.

2.1. Fingerprinting-based detection model

The detection model we work on in this paper is described by Wi-Fi client devices entering the effective observation range of a listening Wi-Fi device such as a router, which may collect and share observed data from these devices. Only the received packet data is considered, without taking into account the device's distances based on received signal strength or their precise geographic location. The goal is to separate devices from each other and identifying vendor and model by analysing captured Wi-Fi packets sent by these devices and received by a listening device, while omitting the MAC addresses from the analysis. The listening device is managed by the fingerprinting system, while the target devices are non-cooperative.

2.2. Probe requests

In general, we cannot expect a non-cooperative device to connect to a nearby Wi-Fi network while we are listening in. Therefore, we can only observe frames that are exchanged without being associated to a network, e.g. for network discovery (so-called Class 1 frames [5]). In this category, PRs are most promising for DI because they are transmitted consistently and frequently. PRs are widely used in related research: They facilitate network discovery and exchange of basic parameters

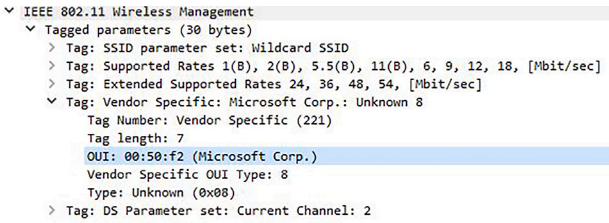

Fig. 1. An example for IEs (“Tagged parameters”) in a Wi-Fi PR taken from the UJI data set, rendered in Wireshark. Shown are a few possible IEs with values listed for the tag: “Vendor Specific”.

for network connection establishment. This requires a lot of information about both parties' capabilities, configurations and characteristics, which can also be used for identification. PRs contain Wi-Fi MAC addresses for both the sending and receiving device, which are used for device addressing in shared transmission mediums. The seventh bit of the first byte denotes whether a MAC address is locally (if set to 1) or globally administered (if set to 0) [6], which we denote as the *global bit*.

A PR further contains a set of IEs, which are used in PRs to signal capabilities and supported features to nearby networks prior to connection establishment. An example IE is depicted in Fig. 1. There are no guidelines or standards on how vendors should implement the usage of IEs, and the IEs used in PRs across vendors vary drastically, as we show in Section 5.2. Since the capabilities and supported features of a particular Wi-Fi device usually do not change often, a device's IEs are relatively stable. PR fingerprinting algorithms leverage the heterogeneity and relative stability of IEs and their values to group PRs that originate from the same device.

2.3. MAC address randomization

MAC address randomization is more widely implemented in modern devices, where it is either optionally enabled or pre-configured by default. Devices that use MAC address randomization for network discovery periodically replace their MAC address with a randomly generated address while sending PRs. This should render the MAC address useless as an identifier for observers. We make use of the *global bit* to decide whether an observed MAC address is randomized or not. This is common practice in related research, as seen in [7].

2.4. Challenges when working with probe request data sets

PR data sets which do not provide ground truths for the sending devices of the captured packets are more widely available and larger in scale, as the collection of a ground truth is a challenging endeavour. For these data sets, deriving vendor and/or model of a device can be done by running the observed MAC address through an Organizationally Unique Identifier (OUI) lookup table. These do not work on MAC addresses that randomize the OUI of the MAC address. Also identifying the vendor of a device by setting the correct OUI is not required, and also would not identify the device model either. Both these aspects make the use of an OUI lookup table an unreliable source to establish a ground truth on an unlabelled PR data set.

Though randomized MAC addresses are always locally administered, not all MAC randomization implementations correctly set the *global bit*. For example, some implementations fix the global bit to 1 and randomize the remaining 47 bits of the MAC address, others randomize all 48 bits. Again other implementations leave the vendor address bytes (OUI) intact and randomize only the interface-specific bytes of the MAC address. Thus, using the *global bit* is an unreliable way to determine the usage of a randomized MAC address in a PR. This poses further problems when working with unlabelled data sets of PRs and renders the OUI lookup table an even greater liability to set a ground truth.

Additionally, Fenske et al. postulate that PRs originating from the same device may adjust the contained IEs depending on the MAC address randomization setting chosen [1]. Since the majority of the following analysis is performed on PRs with randomized MAC addresses that a ground truth is available for, we expect that this mechanism of IE adjustment does not impact our findings.

Finally, devices can work in different Wi-Fi operating states, some of which can be triggered by the user, such as sleep, power save, screen locked, screen enabled, Wi-Fi screen in use and possibly more. Previous work suggests that the operating state can have an influence on the intervals between as well as the transmitted information by sent PRs. Many data sets do not label the operating state of a device, from which PRs were captured from. If these states impact the content and structure of a received PR, not having access to the operating state makes understanding their usage much more difficult, highlighting the importance of having states labelled in data sets. On the other hand, an attacker would likely never have an information on the operating state of a device to be tracked, if that device does not communicate its state. Understanding the impact of operating states of Wi-Fi devices may allow to use observed device behaviour in different states to identify devices that otherwise would look identical to an observer.

2.5. Related work

We summarize research on the usage of IEs for DI, device counting and data sets of PRs with and without MAC randomization. The use of IEs for DI and PR fingerprinting is a fruitful research field. Vanhoef et al. [3] study a fingerprinting approach based on the heterogeneity of IEs to cluster devices based on their PR content rather than on the MAC address. While their work lays the ground work for many years of research to come, they base their investigation on data that is by now over 10 years old and does not account for modern advancements in both the Wi-Fi standard as well as modern devices and their capabilities. In particular, the used data does feature only very little PRs with MAC address randomization enabled. However, they conclude that MAC address randomization does not prevent DI under their investigation, which is a conclusion that should not be drawn given the data used and knowledge established in Section 2.4. We evaluate their approach on modern devices and improve on it. Their basic approach does not hold up well on modern devices, as we show in Section 5.4.

Fenske et al. [1] base their work on Vanhoef's idea and investigate their applicability on modern devices. They conclude that modern implementations of MAC address randomization largely succeeds in protecting privacy, as some problematic features and IEs like Wi-Fi Protected Setup (WPS) were no longer in wide-spread use. Even though their investigation sheds light on various different aspects, further research shown below indicates that MAC address randomization on modern devices still does not protect against all kinds of analysis.

Recent approaches like [2,8–10] demonstrate that clustering and machine learning can be a useful tool in separating devices i.e. for the purpose of device counting, though we notice a lack of work on identifying *individual* devices. Information on some content and specifically the structure of PRs and IEs gets lost or underrepresented when relying on automated classifiers such as random forest or feature extraction such as Principal Component Analysis (PCA). These recent approaches also do not set a focus on perfect device separation, where small numbers of misidentified PRs are tolerated towards an average score across all observed packets. This might be acceptable for device counting operations, where small standard deviations can be tolerated. Given a law enforcement centric use case, a misidentified device can lead to a wrongful (non-)conviction, which is why a perfect separation of PRs is worked towards.

Extended approaches such as Mishra et al. [11] combine content-based attributes from PRs additionally with time-based features (e.g., inter-frame spacing) to associate randomized MAC addresses belonging to the same device. Their approach leverages the presence of the eight most frequent IEs for signature construction, but does not consider their order or arrangement within the frame.

Over the years, various PR data sets with varying scope, quality and labelling were published. Bravenec et al. [7] compare six data sets, three of which we use our investigation in addition to our own labelled data set (see Section 4). Published by Bravenec et al. [12] in 2023, the **UJI** data set compiles ∼1.5 million unlabelled PRs, a large amount of which use MAC address randomization. As this is the most recent data set following a comprehensible capturing methodology and sports a decent size, we included this data set. In 2022, Pintor et al. [13] published the **Cagliari** data set. It contains labelled measurements of a small amount of devices, using MAC address randomization and different capturing modes. This is the only data set we found that contains labelled PRs from devices using MAC address randomization, which is important to our investigation, so we included this data set. The **Sapienza** data set [14] was published in 2013, before the market success of MAC address randomization. However, this data set has been used for evaluation by multiple researchers, which allows comparison with our findings. It also contains over 11 million packets and enables the investigation of the evolution of PR templates, so we included this data set.

3. Methodology

The goal of our methodology is to separate devices from each other, given sets of observed PRs by analysing and evaluating only their content. We consider a device uniquely separated if, after using our DI approach, all PRs sent by that device are attributed towards it and no PR sent by different devices are attributed towards it. To that end, we study how much information is carried by the structure of PRs. We do this by deriving so-called templates from the PRs and using these to split the large set of observations into smaller subsets, which help in identifying separating IEs that assist in labelling. An attribution algorithm may then run on smaller sets of PRs, which may allow it to work on a higher dimensionality when analysing the available data.

Previous work on PR fingerprinting uses various feature extraction techniques (e.g. PCA) and classification methods (e.g. random forest) to identify or classify a device and determine the IEs with the highest overall score for successful clustering of PR sets [2,8]. Doing this reduces the overall complexity of the data set while keeping those IEs providing information deemed to have the most overall value towards clustering. As noted in Section 2.2, PRs may contain a plethora of different IEs, with most individual PRs being sparsely populated, which our findings support (cf. Section 5.2). This approach trades seemingly marginal information loss for a more manageable calculation complexity and processing overhead. While some DI and device counting approaches show that reasonable accuracy can be achieved using such feature reduction (as discussed in Section 2.5), it is possible that uncommon IEs may assist in distinguishing otherwise identical PRs.

During our investigation we observed some fascinating anecdotal evidence that supports the notion of not dropping information for DI, but rather find ways to improve the performance of identification systems, e.g. through data set compression using filtering. These examples are illustrated in Section 5.2.

3.1. Construction and usage of probe request templates

Our approach uses PR templates to conserve structural information such as existence, order, and amount of all IEs and leverage this information to group PRs by template. Using PR templates as filters results in a meaningful compression of the data set when looking for specific subsets such as all PRs of a particular device. *PR templates* (from here on templates for short) are constructed as follows:

Table 1
Examples of PR FPs generated by different selected devices with the methodology by Vanhoef. The FP is a textual representation of a subset of IEs and their content.

Device	FP
Device1	*0,1,50,supra:02040b160c121824, extrates:3048606c*
Device2	*0,1,50,3,45,127,supra:02040b16, extrates:0c1218243048606c, htcap:002d, htmcs:000000000000000000000000000000ff, htext:0000,httx:00000000,htasel:00,htampdu:17*
Device3	*0,1,50,221_0050f2_08,supra:82848b960c121824, extrates:3048606c,221_0050f2_08:001500*

Table 2
Examples of different templates generated for the same selected devices as in Table 1.

Device	Template
Device1	*tags;supra;extrates*
Device2	*tags;supra;extrates;htcap;htmcs; htext;httx;htasel;htampdu*
Device3	*tags;supra;extrates;221_0050f2_08*

First, for a given PR its PR template is generated by collapsing all IE tags to the string *tags* and concatenating the labels of the other IEs without their values, with examples shown in Table 2. This way, a PR template is an abstract of a given PR only denoting order and existence of IEs.

Next, the PR FP is calculated based on its template using an adopted version of the fingerprinting method proposed by Vanhoef et al. [3], where **all present IEs** are used and their values are evaluated. Examples for full fingerprints are shown in Table 1. For this step, tag 3 *DS_channel* is used for both the template and the FP but its value is ignored for the FP. We are interested in the existence of tag 3 and its position, but since devices probe on various channels, tag 3 would change the FP if their values were considered for fingerprinting while not necessarily indicating different devices. Service Set Identifier (SSID) values are omitted as we worked on pseudonymized data only. When SSID information is available, evaluating its value should be used to improve precision.

The tags introduced by Vanhoef et al. [3] served as a starting point and were subsequently extended by going through the data sets, checking for new tags and adding them to the list. For new tags, we chose expressive but concise string representations to convert binary data into a FP. Vendor specific tags (221) and extended tags (255) are incorporated by using their vendor OUI, vendor type or extended subtype to form `221_<vendor-OUI>_<type>` and `255_<sub_type>` respectively. A list of tags in use for IEs in this study is given in Section 4.2.

To group PRs by templates, the following steps are performed for each device (by MAC address for unlabelled data sets or by label for labelled data sets). A pseudocode for this procedure is also given in Appendix A:

1. For each PR of a given device: calculate FP
2. For each FP: calculate template
3. If the template is new: create a set for FPs with that template
4. add the FP to the set with the corresponding template

Using this logic, all PRs with the same template will be grouped in the same set and by extension, all devices with stable templates (i.e. devices that generate only one template over all their PRs) have added all their PRs to a single set. The template acts as a filter for the entire data set. Devices that do not have stable templates require some additional work. Generally speaking, a device generates more than one template because the IEs or their order change between PRs sent within the observation period. Filtering for only one of its generated templates will fail to filter all of its sent PRs. We introduce handling of these cases in Section 3.3 and discuss examples in Section 5.4.

Using templates as data set filters allows to filter out PRs that – by structure – cannot belong to the target device. This filtered partition comprises structurally identical PRs, which allows clustering approaches to now work on a more homogeneous data basis of smaller size. Using more features to enhance the identification precision is enabled by this data set filtering method. However, DI using trained algorithms would now require training the algorithm for each different template for a data set encompassing solution. If only ever one target device (see law-enforcement use case) needs to be tracked, a trained algorithm for that particular device would be sufficient.

We refer to the rate of the relative size of the filtered partition to the overall data set size as the compression rate. Though we hypothesize that using templates as a compression mechanism can have a positive effect on trained classifiers for specific templates, since we do not use trained classifiers in this work, we cannot prove this hypothesis. The DI accuracy of our presented approach is not impacted by the compression rate, as our knowledge base approach inherently utilizes templates to structure the knowledge used for the DI algorithm. Rather, its hardware size and time requirements are impacted, as the knowledge base (cf. Section 3.2) stores device attributes by template and only stores differences under a template, rather than the entire fingerprints per device. We did not evaluate the overall expected storage and time improvement using templates for the knowledge base.

To further characterize our templating method, we hypothesize that templates are

- **stable per device**, meaning that all PRs sent by a single device generate the same template,
- **diverse**, meaning that there are many different templates that packets generate,
- contributing to a **significant compression** of the data set, meaning that using a template to filter a data set will filter out a great majority of packets,
- **evolving**, meaning that the evolution of the capabilities of Wi-Fi devices and their features have an impact on the performance of templates.

To test these hypotheses we use four different PR data sets as described in Section 4. We aim to develop a greater understanding of how the structure and distribution of IEs can assist in MAC-address-agnostic DI.

3.2. Constructing a knowledge base

Using the above filtering method with PR templates a knowledge base can be accumulated using our ground truth for a given data set. Using that knowledge base, an optimized DI approach is derived. An excerpt of the knowledge base can be found in Appendix A. The knowledge base is constructed as follows:

First, calculate all templates for all PRs in that data set and group the FPs as described above while maintaining ground truth information. For each device, we check how many templates its PRs generate. Then we check whether these templates are unique to this device or are shared with other devices. If the templates are unique, we assign the set of templates to this device. Now, this device can be uniquely identified within the data set, if a PR is observed that would be grouped by either template of that device.

Else, if at least one of the templates is shared with another device, both (or more) devices sent at least one PR with that template. In this case, we must resort to using the FP to distinguish the devices. As there may be a large number of FPs, the following process is performed to reduce the template to those IEs that differ between devices:

1. If the FPs of PRs are identical, stop. In this case, the devices are inseparable based on their PR FP. Save this template for all devices that share it.
2. Else: Find a separator - aggregate IEs as part of this template, where the FPs differ in value.
3. Identify a set of these IEs that perfectly separate all devices that share this template as a templating rule.
4. Check whether that templating rule already has a set of FPs assigned to it. If yes, repeat Step 3 with another combination, if possible. Else save the templating rule for the corresponding devices.

This approach allows to save only the differing parts for a fingerprint under a template. The template reduction in step 3 aims to streamline these reduced FPs for a devices with unstable templates, of which some are shared with other devices, in order to reduce the amount of FPs that need to be saved.

3.3. Device identification using templates

Once all PRs have been grouped by templates, the data set consists of a collection of filtered PRs sets. Each of these sets only contains PRs matching a single template and is smaller in size than the original data set of PRs, assuming more than one template was found in the data set. We compare the overall performance of a novel DI approach on these reduced data sets to previously proposed DI approaches.

The fingerprinting approach proposed by Vanhoef et al. [3] is used first as it is optimized for data collected before 2016. Although this approach was designed for older data sets, it could prove effective, especially when applied to the Sapienza data set [14], published in 2013, which we use for comparison. We compare the features selected in their paper to our templates and the knowledge base we gather on IEs over newer data sets to evaluate the evolution of IEs since that publication. Furthermore, we compare our templates to the features selected in recently published papers that predominantly use clustering on a lower feature count with the goal of device counting. Compared to our previous work [15], the following adaptation was made possible due to a significantly more efficient implementation written in Rust, as the python library `scapy` was very limiting in the amount of tests over time. A pseudocode for our algorithm is provided in Appendix A.

For each PR, its template is calculated. The knowledge base is consulted to see, which devices use that template. Using the templating rules saved for this template, the reduced FP is calculated and compared to the knowledge base. The PR can then either be attributed to a specific device, if the reduced FP matches, or be attributed to a bin, that may have other devices share it during further examination of a given observation. Using the knowledge base allows to assign PRs with originally different FPs to the correct devices, which establishes a template-based link between FPs. If two devices share the same FP before template reduction, they will also share the same reduced FP. Devices sharing FPs are indistinguishable by our approach, resulting in an imperfect separation of the sharing devices for the performance evaluation of the approach. In that case, other information such as a modern approach towards sequence numbers or temporal information or Received Signal Strength Indicator (RSSI) or a behavioural analysis (cf. 5.3) could be utilized, which we did not investigate the performance for. Since we did not study any way to leverage information from outside the raw PRs, instead a tiebreaker votes all PRs sharing the same FP towards a device that is most likely to use this observed FP. This tiebreaking method is flawed in many ways, as it favours more talkative devices or simply devices of which more PRs were collected for the data set. Still, this method serves as a simple non-breaking termination to the algorithm, which will need improvements in the future. In some cases, this tiebreaker was infeasible, for too many devices shared fingerprints or their was no clear "most likely" candidate. In those cases we either attributed all these FPs to a `Generic Template` or a template named after two devices. Though for our metrics, these shared templates will fail all devices for the purpose of DI accuracy, we felt it held more value to highlight the overlapping devices rather than having our algorithm guess.

The results are then compared to the ground truth to evaluate the identification performance of the approach. The performance is then compared to the baseline of the Vanhoef fingerprinting performance. Since the PRs are attributed to bins, which express an attribution to a single device (for the most part), we can use these to also compare our results to Pintor et al. [8] in device counting accuracy. Finally, the approach described above features a comprehensible set of operations, leading to explainable results.

4. Data sets

With our methodology stated, next we will describe the data sets we worked on and how they were pre-processed to fit our requirements. The four data sets used are shown in Table 5, where the last one labelled "Bonn" was captured by us. For our investigation, all data sets were pre-processed to satisfy some requirements.

4.1. Existing PR data sets

For the unlabelled data sets (**Sapienza** [14], **UJI** [12]), we removed all packets that had their MAC address randomized (see Section 2.3). Due to lack of explicit labelling, there is no ground truth available for these packets. For packets with non-randomized addresses, the MAC address is used as a ground truth to label PRs as coming from the same device. For UJI, we also noticed that 772 MediaTek MAC addresses (OUI `00-0c-e7`) make up 31.67 % of all MAC addresses in the data set, but only 5271 PRs or 0.56 % of all PRs. Based on the consistency of the structure and the timing of these PRs, we concluded that these 772 MAC addresses originated from the same device. More specifically, each of those MediaTek MAC addresses was used in a distinct time period without overlapping with the use of other MediaTek MAC addresses. As such, we deemed these PRs as having been sent by the same device and labelled them accordingly.

Table 3 shows an analysis of device vendors present in the UJI data set and the labelled data sets described below. An OUI lookup was utilized to derive vendors of specific devices, as no ground truth was given. 562 out of 1666 devices or 33.73 % were matched to either belonging to Apple, Google, Huawei, Motorola, OnePlus, Samsung or Xiaomi. For why only a third of all observed devices were attributed to these ubiquitous vendors, for which would expect a larger market share, we refer to our discussion in Section 2.4.

As shown by the MediaTek anomaly, we cannot definitively conclude that all randomized MAC addresses in the UJI data set have been removed based on the *global bit*. Thus, the performed vendor analysis can only derive vendors from OUIs gathered from non-randomized MAC addresses or from randomized MAC addresses that happened to be interpretable as a valid OUI. We are aware that this analysis requires reliable removal of PRs with randomized MAC addresses. Still, given the missing labelling, this is the closest estimation we can provide.

Since both the Sapienza and UJI data sets do not contain reliable ground truth information for PRs with randomized MAC addresses, two other data sets are used. Both labelled data sets are much smaller in size because the recording and labelling required to compile such a data set is very time-consuming.

Published in 2022, the **Cagliari data set** [13] compiles PRs with randomized MAC addresses labelled per device and capturing mode. For this data set, we used the labels provided by the data set to split the data by device and mode. Modes labelled by the data set comprise "screen active/inactive", "Wi-Fi on/off", and "Power saving on/off".

Table 3
Devices by vendor for the UJI, Cagliari, and Bonn data sets. The UJI information is derived from OUI lookups of only non-randomized MAC addresses. For Cagliari and Bonn a ground truth is known.

vendor by data set	UJI	Cagliari	Bonn
Apple	80	5	3
Google	6	2[a]	2
Huawei	71	4	–
Motorola	47	–	3
OnePlus	2	1	1
Samsung	147	5	3
Xiaomi	209	5	3
other	1104	–	–

[a] Cagliari has two Google Pixel 3A phones. No other multiples of specific models are known in the data sets.

We then grouped the PRs for each device/mode for all Wi-Fi channels, as we do not consider channels. Data for 22 different devices was collected, as shown in Table 3.

When examining this data set we noticed a lot of problems that may negatively impact our results. Several device traces were captured in an anechoic chamber, while others were not, but none were captured in both. This data set has the only instance of measurements from two different devices sharing vendor and model (Google Pixel 3 A) where one measurement was done in an anechoic chamber and for the other measurement no information about the environment is given. Having a data set captured in both noisy and noise-free scenarios provides a huge opportunity to evaluate whether a DI approach is robust to these conditions. Additionally, not all devices in the data set are measured in all modes. We chose to include all traces from both scenarios for our evaluation, as we expect these scenarios to have little impact on our approach. The performance of our DI approach was evaluated on all device modes combined, which we refer to as the *merged mode*.

4.2. New PR data set bonn

As the Cagliari data set [13] does not hold up well to in-depth scrutiny, we decided to compile a **Bonn PR data set** with randomized MAC addresses labelled per device. In particular, we found the claim that devices change the content of their PRs when the MAC randomization mode is changed to not be investigated yet. The data set is published and will be continuously updated with additional measurements.[1] We aim to significantly expand the data set through automation and crowdsourcing as described in Section 6. Our goal is to compile a data set of measurements labelled according to a common list of labelling rules. This data set should collect measurements for a diverse set of parameters and devices, but in particular measurements where only one parameter differs. At the time of publication, the currently collected measurements are still lacking certain combinations of parameters per device, which we criticized the Cagliari data set for. However, every entry is completely labelled and pseudonymized. As described in Section 6, plans to fill gaps and expand measurements, while maintaining quality, have already been developed but are still to prove their worth in practice.

Currently, the data set consists of the following 15 devices: iPhone 11, iPhone 13, Xiaomi Redmi 9, Moto g23, Moto g5s, and a Samsung a51 (already present in the WiMob paper [15]) and iPhone 16, Pixel 9, Pixel 7a, OnePlus 12, Motorola Moto G Stylus 5G, Samsung Galaxy A15, Samsung Galaxy S24, Xiaomi Redmi 13C, Xiaomi POCOPHONE F1 (new). The data for the new devices was measured using the automated system described in Section 6.1, whereas the other devices

[1] Probe request data set will be updated over time and is available at https://doi.org/10.60507/FK2/TXOWTX.

Table 4
List of all dissected Tags in IEs across all examined data sets with their corresponding value strings used in our templating. VH: Vanhoef, Sap: Sapienza, m: Modern data sets: UJI, Cagliari, Bonn.

Tag	Descriptive String	VH	Sap	m
	tags	✓	✓	✓
1	supra	✓	✓	✓
3	DS_channel		✓	✓
10	Req_country_info			✓
45	htcap	✓	✓	✓
	htampdu	✓	✓	✓
	htmcs	✓	✓	✓
	htagg		✓	✓
	htext	✓	✓	✓
	httx	✓	✓	✓
	htasel	✓	✓	✓
50	extrates	✓	✓	✓
59	Supp._Op._Classes_curr			✓
	Supp._Op._Classes_all			✓
70	RM_En_Cap			✓
107	interworking	✓		✓
	interworking_HESSID			✓
114	MeshID			✓
127	extcap	✓	✓	✓
150	unknown		✓	✓
191	VHT_Cap_info		✓	✓
	VHT_Sup_MCS		✓	✓
221	221_<vendor-OUI>_<type>	✓	✓	✓
255	255_<sub_type>			✓

Table 5
Statistics of the used data sets as well as average template performance on those data sets: whether the data set is labelled, number of data points (PRs), publication year, number of devices, number of templates, average templates per device, average devices that share a template, and average compression achieved by templating.

data set	Sapienza	UJI	Cagliari	Bonn
labelled?	✗	✗	✓	✓
#PRs	3,943,700	918,054	69,701	21,333
year	2013	2023	2022	2025
#devices	305	1667	22	15
#templates	20	145	25	20
avg. t/d	1.0	1.08	1.68	1.33
avg. d/t	15.25	12.39	1.48	1.35
avg. comp.	95.0 %	99.0 %	98.0 %	95.0 %

were measured using a manual process described in the previous paper [15]. The measurement configurations listed below are the same for all measurements, but the new automated system uses different monitoring hardware (3x Alfa Networks AWUS036ACM 802.11ac USB adapters instead of 3x 802.11n ESP-32) and automatically sets the correct current device state using ADB where possible.

The new devices were measured inside a shielding box with an attenuation of approximately 60 dB. The old devices were measured in a “random noisy” environment without a shielding box in an office building while placing other Wi-Fi devices at least one metre away. During the measurement each device is placed directly next to or on top of the measurement system, allowing RSSI-based filtering. Both the old and the new devices were measured in the following modes, aiming to have overlapping measurement modes with the Cagliari data set:

1. **Mode AR:** screen on and unlocked, Wi-Fi settings activity open, Wi-Fi connected, MAC randomization disabled, power saving features disabled
2. **Mode S:** screen locked, Wi-Fi disconnected, MAC randomization enabled, power saving features disabled
3. **Mode PS:** screen locked, Wi-Fi disconnected, MAC randomization enabled, power saving features enabled

Before starting the measurement, at least one Wi-Fi network is added to the device’s Preferred Network List (PNL). Some devices (e.g. Xiaomi Redmi 13C) very rarely send out PRs when the PNL is empty. To prevent this, all devices are once connected to a local Wi-Fi access point which is then turned off before starting the measurement. An in-depth analysis of the impact of PNL state on probing behaviour is left for future work.

The new devices for measurement with the automated system were chosen based on a market analysis, considering popularity of the manufacturer and model and date of release. The goal was to choose current devices from best-selling product lines of well-known manufacturers, taking availability into account. This ensures that our analysis is based on devices that are either already very common or can be expected to become popular within the next 1–2 years. While the previous version of our data set contains devices released back in 2019, the new selection therefore focuses on recently released devices. We assume that these newer devices are more likely to have implemented stronger countermeasures against PR fingerprinting. Due to the greater variability between models and manufacturers as well as the easier measurement with the ADB automation, we have focused primarily on Android devices in the new selection.

The underlying selection introduces several biases, which are likely to have an effect on our results, though may only be solved by extending the data set (see Section 6). Since the device selection is based on a market analysis in Germany, our findings may not reflect the performance of our proposed approach on a similar selection based on the analysis of other markets. Furthermore, since we focused on having a broad spectrum of different devices available, we lack insight of our approach’s DI performance for scenarios with multiple very similar devices, like overlapping vendor and model. The focus on Android phones may disproportionately lean our results towards Android specific observations. Having only selected recently released devices, our evaluation cannot conclude how susceptible older devices are to our approach. To remedy these shortcomings, other data sets with more and in case of Sapienza older devices have been selected, though with the above discussed problems. The compilation of the data for all measured devices results in a fourth data sets of PRs, where the labelling allows us to verify whether our DI approach groups PRs and devices correctly together.

Table 4 contains a list of IEs we observed within the examined data sets. The column `VH` denotes the tags introduced by Vanhoef et al. `Sap` the tags present in the Sapienza data set, which Vanhoef et al. used for their examination and `m` denotes the tags observed across

5. Experimental results

First, we present and discuss in-depth template statistics we observed. Second, we summarize the IEs used as features for fingerprinting or clustering, respectively, as presented in related research with the ones we observed in all data sets. We dissect the IE usage per device observed in the used data sets and describe our observations. An investigation follows that helps understanding how the observation time impacts the gathered data per device. Then the results of our DI experiments are presented.

5.1. Template statistics

Using the data sets described in Section 4, we calculate the template for each PR. After calculating all the templates, we determined a number of metrics that measure the performance of our DI approach on the data sets. For each data set, Table 5 shows some basic statistics such as number of PRs after pre-processing (#packets), its publication year (year) and the number of individual Wi-Fi devices based on labelling (#devices). It also shows the results of applying the template filter to the whole data set: The total number of unique templates (#templates), the average number of templates that a single device generates (avg. t/d), the average number of devices that share a single template (avg. d/t) and the average degree of compression achieved by the templating filter (avg. comp.). For a later investigation (cf.

Table 6
Statistics on the number of templates generated per device, both absolute and relative to data set size. For each data set, the number of devices that generate one, two, three, four or five distinct templates is given. For Cagliari and Bonn data sets, the shown amounts represent the results for the merged mode. In the Sapienza data set all devices generate one template and is therefore not listed.

#templates per device	UJI		Cagliari		Bonn	
1	1565	94 %	10	45 %	6	43 %
2	81	5 %	9	41 %	7	50 %
3	15	1 %	3	14 %	–	–
4	4	<1 %	–	–	1	7 %
5	2	<1 %	–	–	–	–

Table 7
Average number of IEs per template per measuring mode per device for the Bonn data set. Devices appear to use more IEs for PRs when in mode AR. No difference was observed between mode S and PS. Numbers with a decimal appear when a device used multiple templates with different lengths.

device by mode	AR	S	PS
iPhone 11	17	–	17
iPhone 13	–	17	17
iPhone 16	21	20.5	20.5
Xiaomi Redmi 9	–	5	5
Xiaomi Redmi 13C	5	5	5
Xiaomi Pocophone F1	19	18	18
Moto g23	5	5	5
Moto g5s	–	7.5	7.5
Moto G Stylus 5G	17	16	16
Samsung A51	–	13	13
Samsung Galaxy A15	5	5	5
Samsung Galaxy S24	24	23	23
Google Pixel 7a	22.75	22	22
Google Pixel 9	–	5	5
OnePlus 12	25	24	24
average # IEs	15.97	15.39	15.39

Section 5.2), aggregating unique templates over both the Cagliari and Bonn data sets yield 42 unique templates. This is due to the fact, that some devices share templates across both data sets, or in other words, three templates are shared between devices of both data sets.

For the Sapienza data set, no individual device generated more than one template, meaning that the templates are perfectly stable for that data set. In Table 6, we can see that for all modern data sets, the average amount of templates generated per device is still close to 1.0, but some devices generated more than one template. From the Cagliari and the Bonn data sets, we know that the state of the observed device can have an effect on the content of its PRs and therefore the corresponding templates. We can see this effect in Table 7, which shows the number of IEs per template in the Bonn data set by device and mode. Some devices generate multiple templates even though their state is not changing. It is not clear what causes these devices to sometimes append IEs that were not used for the last set of PRs. Possible explanations include factors such as the surrounding Wi-Fi environment, nearby networks, medium contention, or other variables. Since PR data sets do not capture the context in which they were captured, we cannot know for certain. However, we can state that the number of IEs present in a PR does not seem directly indicative of the vendor, as all vendors, from which we had access to more than one device, vary significantly with different models.

Fig. 2 shows a detailed view of the number of devices per template for the different data sets. The older data sets (Sapienza and UJI) have some templates that are shared only between a low amount of devices, but also exhibit a number of templates that are shared by many different devices. Widely shared templates not only limit data set compression, they also hint towards multiple devices that produce similarly constructed PRs, which in turn may share FPs and thus may

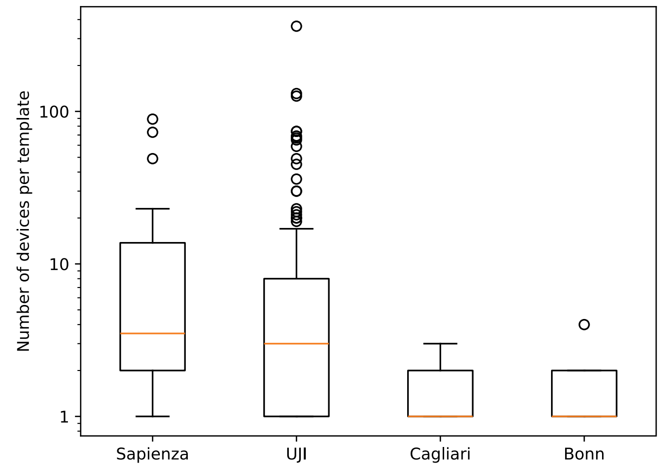

Fig. 2. Box plot of devices that share templates per data set. Less devices sharing a template means that templates are more indicative for individual devices and therefore more useful for device identification. Large data sets (Sapienza and UJI) have a greater amount of templates that seem to be shared by many devices.

be indistinguishable based on packet content alone. The labelled data sets (Cagliari and Bonn) exhibit a consistently low number of devices per template. Given the observations shown in Tables 3 and 5, the size of the data set may have an effect on the amount of shared templates, especially when many devices are observed that stem from the same vendor or manufacturer. Even though UJI and Cagliari have been captured at almost the same year, surrounding factors like data set size, composition, and capturing methodology differ, such that statistical properties are difficult to compare.

For Cagliari and Bonn the device selection is broad, with some repeating vendors but rarely repeating device models. Given that for Bonn this device selection process was done deliberately to maximize the market coverage, we expect the same reasoning for the device selection for the Cagliari data set. UJI does not provide any ground truth as to which devices the PRs were captured from, while Cagliari and Bonn gave that information. According to the authors, the measurement process was performed over a month at a static location, which explains the larger amount of observed devices. The captured devices were not deliberately selected, but simply those used by people that passed by the measuring location. For UJI it can be assumed that the captured devices are biased according to the Spanish phone market at the time of capture. We can assume that, depending on people visiting the university's measurement location, a certain market coverage is achieved with many devices captured sharing vendors and models. Additionally, as discussed in Section 4.1, it is possible that single devices used randomized MAC addresses but evaded the removal by filter, due to not setting the *global bit*. These would be counted as multiple devices, even though their PRs were identical apart from the MAC addresses, leading to shared templates, which could explain the high outliers seen in Fig. 2. Therefore templates that are shared by many different devices are not only more likely to be observed in larger data sets but also in unlabelled data sets where common preprocessing may introduce unknown errors. Unintentionally, PR templating may prove a useful tool for data set cleansing. With the labelled data sets, the smaller variance of devices allows for a consistent and high compression performance.

Based on the average statistics in Table 5 as well as the information in Tables 6 and 7, we can draw the following conclusions regarding our previously established hypotheses: PR templates are **diverse**, and increasingly so for more modern devices. As more and more features are supported by Wi-Fi devices, different IEs are being used, which directly impact the templates that can be derived from these PRs. However, modern devices in fact tend to use multiple different templates for their PRs, depending at least partially on the mode the device is in. Therefore we can conclude that PR templates are **not stable per device** for modern devices. Given the observed diversity of PR templates, the proposed knowledge base may yet allow to separate devices even with unstable templates, as we show in Section 5.4. Also, we can observe that they contribute to **significant compression** for all data sets of at least 95 %. Comparing template statistics between the Sapienza data set and the modern data sets, we can see that the number of devices per template has significantly decreased while the number of templates per device has slightly increased. While a more detailed analysis is necessary to characterize template evolution, this certainly indicates that templates are indeed **evolving** over time.

5.2. Comparison of information element and template usage

The IEs that were used as features for the approaches of Pintor et al. [8] and Uras et al. [2] are now compared to the IEs we dissected and presented in Table 4. When comparing the tag numbers, we observe that most tags are in fact considered by all compared approaches. Tags not shared between all these approaches are tags 10, 50, 59, 70, 114, and 150. Even though the related research does not use all IEs, they use other additional information like sequence numbers or temporal information.

The fingerprinting approach by Vanhoef et al. [3] does not use all of the IEs used by our approach for their investigation on the Sapienza data set, as seen in Table 4. The IE selection by Vanhoef et al. mostly overlaps with the dissected IEs from the Sapienza data set, apart from individual IEs like `DS_channel`, `htagg`, `interworking`, tag 150, and tags 191. When comparing Sapienza with the modern data sets, not using tag 191 Very High Throughput (VHT) does make sense, considering that VHT capabilities saw wide spread support only after 2013 [16]. Omitting tag 3 `DS_channel` also seems sensible, since the varying transmission channels are not indicative of different devices, as discussed earlier. Comparing their IE selection to dissected IEs in modern data sets does show a significant mismatch however, which can explain a poor identification performance, when applying Vanhoef's fingerprinting on modern devices, as we will show in Section 5.4.

Next, an investigation was conducted to understand whether specific IEs are useful for distinguishing between different modern vendors. We omitted the UJI data set for the following analysis, as the missing ground truth disallows any device- or vendor-specific observation or conclusion. Table 8 illustrates the used IEs by vendor in the Cagliari and Bonn data sets. Only the usage of an IEs are taken into account, not their values. **Bold numbers** indicate that a specific IE was used in all templates from all devices of this vendor in that mode. IEs marked with * are only ever used by devices from one vendor. The numbers denote the amount of times, an IE was observed in a template. Some numbers are lower than the device count, as they were not always observed or there was no data for a device in that mode. Conversely, since multiple devices generate more than one template during an observation, some numbers are higher than the device count. The data provided by both the Cagliari and Bonn data sets do not include any PR that does feature IEs with the above mentioned tags 10, 50, 59, 70, or 114, but it does feature one PR with tag 150. We cannot explain the usage of tag 150 in only one PR. Either there was an unknown trigger prompting the device to transmit this singular exceptional PR or perhaps there was a measurement error.

The perfect usage of tags like `supra` and `extrates` was expected. Tag 50 is not in use as a feature in any published work that uses feature reduction on IEs that we are aware of. Yet all available devices utilize tag 50 in their PRs, which could explain, why the feature got dropped through feature selection. Still, we observed iPhones that sent identical PRs but for the value in tag 50, which allowed us to distinguish them. Additionally, exactly one PR was observed that produced a template only comprising `tags;supra`, thus omitting `extrates`. This is the only PR we observed that omitted the tag `extrates` which either is an error while collecting the data or something we cannot explain. Therefor we removed this PR from the further investigation on IE usage. Some IEs always appear together, like IEs belonging to High Throughput (HT) (`htcap`, `htampdu`, `htmcs`, `htagg`, `htext`, `httx`, `htasel`) or High Efficiency (`255_35`, `HE_MAC`, `HE_PHY`, `HE_supp`, `HE_PPE`). These IEs offer no further value past their first instance, apart from when a device actually not uses all of them together for some reason (which we did not observe). Not all vendors use all IEs in the same manner, or even consistently over their individual devices. For example, while the HT IEs are used by all apple devices in all modes, some Google phones use them only in the mode AR but not in PS or S. Of particular interest are the IEs of tag 221 which denote vendor specific information. Though there are three vendor specific IE that are indeed unique to a particular vendor (Apple, Samsung), others are used by multiple vendors, which be counter-intuitive towards the notion that they should be vendor specific.

Finally, the amount of total observed unique templates per vendor per mode is listed at the bottom of Table 8, along with the aggregated amount of PRs by these devices that were examined, which amount to roughly 1628 PRs per device. A total of 42 unique templates are aggregated over both Cagliari and Bonn, meaning that for each template, on average 1357 PRs have been examined. Of those, 18 templates were only observed in less than 20 PRs each, meaning they were rarely shared by multiple devices or modes and were indeed rarely used by these devices that used them at all. Since the used data sets are only of limited size, they only provide little statistical robustness to anomalies such as rare observations or even measurement errors. On the one hand, a rarely seen template may indicate a very strong evidence for a particular device, on the other hand it could also hint towards some error during transmission or reception of a certain PR. Since we have no infallible way to determine whether a rare observation is indeed a correct observation for a device, we gathered statistical data and examined suspicious templates by hand to the best of our ability, with examples discussed in Sections 5.3 and 5.4.

In summary, rarely are IEs vendor specific or even device specific. Their usage however is heterogeneous within the available data sets and can help identifying devices even before taking their values into account.

5.3. Comparison of required observation lengths and device behaviour

To understand for how long a device must be observed to gather as much information about it as possible, we performed another analysis of the labelled data sets considering the first instance of each newly observed unique template and FP. Table 9 shows an analysis of the required time and number of packets for each vendor to include all templates and FP. A 0 denotes that after the first observed packet no different templates or FPs were gathered from further observed packets from the same device. For the latest packets per device mode that introduced newly observed templates or FPs the elapsed times since the start of the observation are denoted in the "after" columns.

We found, that from the examined 34 devices, for 26 (AR), 26(PS), 30(S) devices all observed templates were seen within less than one minute, where for 3(AR), 5(PS), 3(S) devices never before seen templates were observed after more than five minutes. For unique FPs, a similar distribution was found: For 26(AR), 23(PS), 29(S) devices all FPs were observed in less than a minute, where for 4(AR), 6(PS), 4(S) devices unique FPs were observed after five minutes of observation.

Fig. 3 shows the observation results in detail on the example of mode AR. Figures for modes PS and S can be found in Appendix B. For each device the span of observed PRs ranged between two and 20 min, apart from the Samsung S24, which we only have found one PR during a 20 min measurement. For most devices, an observation time of less than a minute is sufficient in gathering all observable information, that was gathered for the entire observation length, regardless of the mode.

Table 8

Number of occurrences of observed IEs per vendor and mode in the Cagliari and Bonn data sets. Bold numbers denote that this IE was observed in every template generated by devices from this vendor.

vendor (num. devices)		Apple (8)			Google (3)			Huawei (4)			Motorola (3)			OnePlus (2)			Samsung (8)			Xiaomi (7)		
IE tag	tag #	AR	PS	S	AR	PS	S	AR	PS	S	AR	PS	S	AR	PS	S	AR	PS	S	AR	PS	S
supra	1	**6**	**8**	**9**	**9**	**3**	**4**	**4**	**3**	**4**	**2**	**4**	**4**	**2**	**2**	**2**	**10**	**6**	**9**	**7**	**12**	**14**
extrates	50	**6**	**8**	**9**	**9**	**3**	**4**	**4**	**3**	**4**	**2**	**4**	**4**	**2**	**2**	**2**	**10**	**6**	**9**	**7**	**12**	**14**
DS_channel	3	**6**	**8**	**9**	8	**3**	**4**	**4**	**3**	**4**	**2**	3	3	**2**	**2**	**2**	7	**6**	8	6	9	11
htcap	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
htampdu	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
htmcs	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
htagg	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
htext	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
httx	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
htasel	45	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	2	2	**2**	**2**	**2**	7	**6**	1	6	8	10
extcap	35	**6**	**8**	**9**	8	1	1	**4**	**3**	**4**	1	1	1	**2**	**2**	**2**	7	**6**	7	4	7	9
255_35	255	3	4	5	6	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0
HE_MAC	35	3	4	5	6	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0
HE_PHY	35	3	4	5	6	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0
HE_supp	35	3	4	5	6	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0
HE_PPE	35	3	4	5	6	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0
255_108	255	1	2	2	2	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
VHT_Cap_info	191	0	0	0	5	1	1	0	0	0	1	1	1	**2**	**2**	**2**	1	1	1	3	7	9
VHT_Sup_MCS	191	0	0	0	5	1	1	0	0	0	1	1	1	**2**	**2**	**2**	1	1	1	3	7	9
255_2	255	0	0	0	8	1	1	0	0	0	0	0	0	**2**	**2**	**2**	1	1	1	3	7	9
FILS_Req	2	0	0	0	8	1	1	0	0	0	0	0	0	**2**	**2**	**2**	1	1	1	3	7	9
interworking	107	1	2	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0
interworking_HESSID	107	1	2	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0
150*	150	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
RM_En_Cap[a]	70	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
221_0017f2_0a[a]	221	1	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
221_0050f2_08	221	1	3	3	7	2	3	3	**3**	3	**2**	3	3	**2**	**2**	**2**	7	5	8	6	9	11
221_001018_02	221	1	3	3	2	0	0	3	**3**	3	0	0	0	0	0	0	4	3	4	0	0	0
221_506f9a_10[a]	221	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
221_506f9a_16	221	0	0	0	5	1	1	0	0	0	1	1	1	**2**	**2**	**2**	1	1	1	3	6	8
221_506f9a_	221	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
221_8cfdf0_01	221	0	0	0	2	0	0	0	0	0	1	0	0	**2**	0	0	1	0	0	3	3	3
221_00904c_04	221	0	0	0	2	0	0	2	2	2	0	0	0	0	0	0	2	1	2	0	0	0
221_00904c_33[a]	221	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0
# templates		6	8	9	9	3	4	4	3	4	2	4	4	2	2	2	10	6	9	7	12	14
# PRs		801	1831	2780	1764	145	287	2186	1801	2219	1185	262	320	1152	436	349	4541	4253	4663	19 054	4089	2871

[a] Denotes an IE unique to one vendor. Also shown are the number of templates and FPs per mode per vendor.

Table 9

Required observed packets and time to observe all templates and FPs in the Bonn and Cagliari data sets grouped by vendor (devices per vendor in brackets) and device modes (AR, PS, S). The absolute numbers refer to the number of packets that must be observed until all unique FPs and templates, respectively, have been observed. The time refers to the amount of time the device must be observed until all unique FPs and templates have been observed.

Vendor (device count)	req. packets in mode AR			req. packets in mode PS			req. packets in mode S		
latest observed new	Template	FP	after	Template	FP	after	Template	FP	after
Apple (8)	0	10	17 s	14	14	5.23 min	6	9	5.57 min
Google (3)	359	1035	7.25 min	0	0	0 s	12	12	5 s
Huawei (4)	0	0	0 s	0	218	6.20 min	0	0	0 s
Motorola (3)	0	0	0 s	47	47	23.38 min	49	49	29.12 min
OnePlus (2)	0	0	0 s	0	0	0 s	0	0	0 s
Samsung (8)	205	205	17.44 min	0	0	0 s	1	2	0 s
Xiaomi (7)	2664	2664	4.57 min	146	146	15.05 min	424	424	15.14 min

This can be important for use cases where a specific target device may only be present for a limited duration to have their packets observed.

Some devices disclose additional information only after many more minutes. It is unclear yet, if and what kind of triggers exist to make these devices send additional info only after a certain period. Possible explanations are external stimuli, like received beacon frames, or internal time-outs, which trigger the device to wake-up, sleep, or change the contents of their PRs.

To understand this observation, we took a closer look into the traces of some outlier devices. Notable devices are the Motorola G5s, the Redmi Note 7, Google Pixel 3 A. In modes PS and S PRs with new information were sent after more than 20 min (Motorola G5s) or 13 min (Redmi Note 7) of being observed. In mode AR new information was sent after over 7 min (Google Pixel 3 A) or even over 17 min (Samsung Galaxy J6). For the first 20 min, the Motorola G5s routinely sends bursts of five to six PRs every 40–180 s, each having their MAC address randomized. However, after 20 min a different set of PRs is being sent sometimes, while continuing with the formerly used routine bursts meanwhile. This new set comprises 16–17 PRs with more IEs, thus a different template and FP, and their factory MAC address. The observations that these different PRs make for 20–24 percent of overall observed PRs from this device and that they follow regular pattern as well as being observed in multiple measurements suggest that these PRs are not a random occurrence. The same behaviour is shown by the Redmi Note 7 in mode PS, though it does send regular bursts of nine to twelve PRs, which only randomize the used MAC address after each burst. After 13 min, it sends a block of 42 PRs using the same MAC address, alternatingly being directed towards a specific SSID or

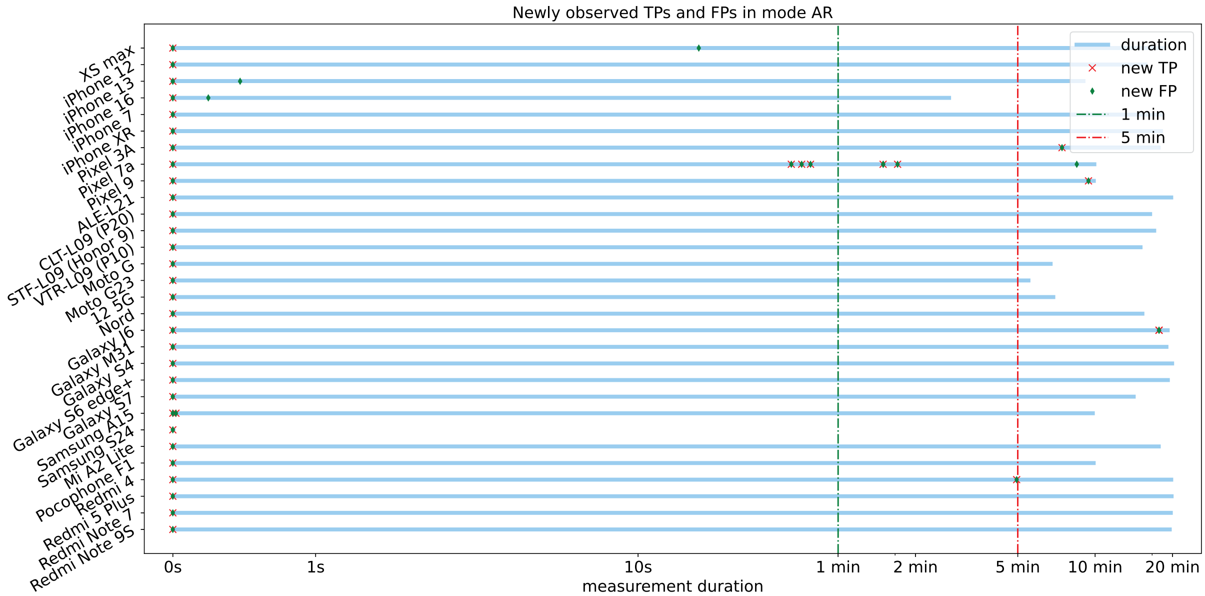

Fig. 3. Timings of newly observed templates (TP) and fingerprints (FP) relative to the overall measurement duration for various different devices. Only four devices showed new information past five minutes of observation time. For the Samsung S24 only one PR was observed.

being non-directed. Since the Cagliari data set does not provide the factory MAC address as a ground truth for this device, we cannot verify, whether the hardware MAC address was used for these PRs as well. This block of PRs makes up around 17 percent of all observed PRs by that device. A very similar block of PRs was observed having been sent by the Google Pixel 3 A in AR mode. The same observation was made for the Samsung Galaxy J6, which directed all 14 PRs of the block of new PRs towards a single SSID. From our filtered data set though, we cannot reconstruct, whether there were i.e. beacon frames received that served as stimuli for this behaviour.

Other observations for devices, that new information was observed from after five minutes of measurement include additional IEs such as vendor tags or HT capabilities, additional extended capabilities octets or bit flips to denote additional HT capabilities. We do not currently understand what causes these devices to change their PRs. Since it is unlikely, that a device's capabilities suddenly change mid-measurement, we expect, that these changes hint towards those devices adapting their PRs in order to signal their capabilities to satisfy broadcast nearby network's requirements.

This analysis was made possible by the usage of templates and fingerprints and hints towards a complex set of behavioural data that is currently unused for DI but could yield considerable value when understood to a deeper degree. If certain devices can be triggered to send different PRs from their usual PRs through i.e. receiving a certain beacon frame, even devices with the same vendor and model could be distinguished by observing their reaction to specific stimuli. We plan to dive deeper into these analyses in future research.

Another important observation is, that for some devices new information was sent after an amount of minutes, where the collection process for other devices was already finished. It cannot be inferred, that all used measurements did provide all information that any one device can feasibly produce. In particular, when the assumption is correct, that certain stimuli can trigger devices to send PRs that they otherwise would not send, these triggers would need to be studied and introduced into measurements. However, it is also uncertain, how long a measurement needs to be, in order to observe all information from a "non-triggered" device. Still, the made observations hold considerable value for use cases, where devices cannot be observed for time periods longer than 20 min.

Table 10
Results for the templating experiments. Shown are the percentage of all devices that have all their PRs separated uniquely when applying naive fingerprinting and after optimizing the DI using expert knowledge.

data set	Sapienza	UJI	Cagliari	Bonn
separation naive FP	40 %	56 %	52.4 %	23.1 %
sep. optimized FP	40 %	75 %	81 %	84.6 %

5.4. Results for the device identification experiments

To evaluate our approach, we applied both our DI approach described in Section 3.3 and the approach presented by Vanhoef et al. [3] to our data sets. In Vanhoef's fingerprinting method, we omit the sequence number analysis described in the original paper. It was used to improve the fingerprinting robustness against devices sharing FPs. Modern devices randomize sequence numbers [1], such that the proposed analysis will instead separate PR which belong to the same device, which is why the analysis is omitted. We refer to the Vanhoef fingerprinting as *naive fingerprinting*. For our approach, which we will refer to as the *optimized fingerprinting*, we extracted knowledge from the templates and FPs as described in Section 3.3 to find optimized fingerprinting features for each template to separate devices using our knowledge base.

Our results are summarized in Table 10. We consider the fingerprinting accurate only if all PR sent by a device are correctly grouped with that device. Our results show that, for the UJI data set, 56 % of devices were accurately identified by naive fingerprinting, which increases to 75 % with the optimized fingerprinting mechanism. Since the UJI data set does not provide a ground truth that can be leveraged to gauge any affiliation between PRs and vendor or models reliably, the UJI fingerprinting was done separately from the Cagliari and Bonn data sets. The Cagliari and Bonn data sets provide a ground truth, which allows to set up a knowledge base as described above and be evaluated together and separately based on the shared knowledge gathered from both data sets.

A particularly striking improvement is shown on the Bonn data set, where an improvement from 23.1 % (or 3/13 devices) to 84.6 % (11/13 devices) is observable. The algorithm only fails to correctly identify PRs by the iPhone 13 and the Motorola G5s. Tables 12, 13, 14 show the results for when the measurement modes are evaluated independently from one another. The Redmi 9 phone needed to be omitted from the evaluation, as we found many packets to be malformed, which hints towards an error during measurement. Devices that could not be separated correctly by the optimized fingerprinting shared FPs with other devices. In case of iPhone 13, our algorithm found both the iPhone 12 and iPhone 13 to be using overlapping PRs in modes PS and AR, thus the tiebreak method (cf. 3) was assigning PRs sent by the iPhone 13 towards iPhone 12 in mode S. We identified a template, that only uses three IEs and is shared by multiple devices, which we call *Generic Template* (cf. A). Devices that us the *Generic Template* are set up to provide the minimal required IEs to satisfy the standard for PRs. Using the *Generic Template* allows the device to probe for nearby networks while keeping most of its capabilities secret, which may hinder their effort to discover networks that require stations to have certain capabilities for a connection. The only devices we observed that use the *Generic Template* were the Motorola G5s and several Xiaomi models, which were never correctly identified due to the generic nature of the used template. This explains, why the PRs sent by the Motorola G5s were incorrectly split into two bins, even though for the Bonn data set no other device used the *Generic Template*.

For the Cagliari data set, our algorithm improved the identification from 52.4 % (11/21 devices) using the naive fingerprinting to 81 % (17/21 devices). The devices our algorithm failed to properly identify, again, were sharing generic templates, mainly the formerly pointed out Xiaomi phones. Interestingly, the amount of bins, our algorithm attributed devices to was 21 (AR), 18 (PS), and 20 (S). When comparing the amount of bins to the amount of clusters Pintor et al. used in their work, we find very comparable results [8]. These results suggest that our approach is a feasible alternative for device counting, even when data from devices outside the Cagliari data set interfere, as our Cagliari analysis was done with the knowledge base that also comprises the templates from the Bonn data set.

When running the algorithm on both data sets together, 29 out of 34 devices (85.3 %) were identified correctly, meaning that all PRs sent by a particular device was attributed to that device specifically.

To explain the performance improvement when compared to the naive fingerprinting, two factors are to be considered. The naive fingerprinting groups different devices together, which can be separated by IEs that are not included in the naive FPs, which our algorithm includes. Also, devices using multiple FPs but a stable template are incorrectly separated by the naive fingerprinting.

When taking a closer look at vendor statistics, some interesting observations can be made. For the examined data sets, Apple, Google, Huawei, and OnePlus devices can be reliably identified. For Motorola and Samsung only individual devices like the Motorola G5s and the Samsung Galaxy J6 cannot be identified properly for all modes. Particularly problematic for our algorithm were three of the Xiaomi devices, as the Mi A2 lite, the Redmi 4, and the Redmi 5+ share templates and fingerprints not only between each other but also with other vendor's devices, being the Motorola G5s. For these devices, other information could be used to identify those, as explained earlier, i.e. in 5.3. From how the presented algorithm's DI performs on devices of different vendors, we gather inconclusive observations. The device sample size per vendor seems to impact the DI, as the algorithm fails for some devices of Samsung (8 devices) and Xiaomi (7 devices), still it perfectly separates all Apple devices (8). Conversely, the algorithm also fails for a device of Motorola (3 devices). Both the Cagliari and Bonn data sets sample devices from the same seven vendors, though with little overlap between the individual models. It is possible that the observed DI performance per set is directly impacted by individual devices within the data sets. The impact individual devices have on the performance would likely be lessened with a larger sample size. On the other hand, a larger sample size may introduce more overlapping devices, thus result in an overall diminishing performance. Since for the Bonn data set devices were picked that were available and span seven different vendors, there was no intent of picking specifically devices that were likely to use disjunct FPs. We allege that there also was no such intent in the device selection for the Cagliari data set.

Table 11
Methodical differences between different compared DI methods, where recent work groups results from [2,8–10].

DI methods	optimized FP	naive FP	recent work
matches all PRs	✓	✗	✗
uses all IEs	✓	✗	✗
explainable results	✓	✓	✗

To summarize, our approach aims to identify devices based on their sent PRs and we demonstrate its prowess and flaws on different data sets, receiving an improved identification performance over its peers, while being comprehensible and explainable in its results. Table 11 highlights methodical differences between our approach, Vanhoef's fingerprinting and recent work as described in Section 2.5 apart from DI precision. The performance of our approach can likely be increased by introducing non-content information such as RSSI information or behavioural information. Utilizing active attacks in order to stimulate a target device to interact with the system in a cooperative way is another possible improvement and may make some behavioural information more feasible to collect.

Finally, it must be stressed that our approach cannot generalize. Unseen PRs can never be attributed to any device, as there is no trained model in use that allows for generalization. For our approach to achieve a wide range of identifiable devices, the knowledge base needs to have been built with many different PRs from many different devices. Therefor it is important to work on larger, well labelled PR data sets. Once such a large and wide knowledge base has been established, the algorithms performance on newly observed PRs can and should be evaluated. Only then is an estimation on scaling issues, particularly towards shared templates, feasible. Nonetheless, we consider our approach to be an important building block towards a system, that uses both observable content-related and non-content-related information to identify Wi-Fi devices which employ MAC address randomization.

6. Crowdsourcing data sets

As described in Section 4 there is a very limited amount of adequate PR data sets available for our use case. More specifically, we require data sets that contain PRs with randomized MAC addresses that are labelled with a ground truth about which PR originates from which device. Therefore we started creating our own data set which would fit our needs. It is labelled in a way that enables us to evaluate our approach, but also allows other possible research using the PRs and metadata contained within. We found our measurement setup to be relatively incomplex and inexpensively reproducible, which facilitated seamless collaboration among the authors of this paper. Given the need for a well labelled and sufficiently large PR data set in combination with a relatively cheap and simple measurement setup, a logical conclusion is to enrich our data set with crowdsourced measurements (Crowdsourced Network Measurements (CNMs)) and crowdsourced packet data generation.

CNMs and the crowdsourced generation of data sets offers a number of benefits in our context: A larger number of contributors increases the data set's coverage to more different devices, models and manufacturers, resulting in a scope that far exceeds what a single contributor could provide. It also offers a means to continually update the data set with new devices as they are released. Lastly, collecting PRs across multiple measurements by different contributors in different environments and configurations allows for observation of behavioural differences depending on parameters like operating system version, geographic region, measurement environment etc.

To ensure that our data set addresses the central issue with other data sets, the lack of a ground truth, we choose to accept contributions

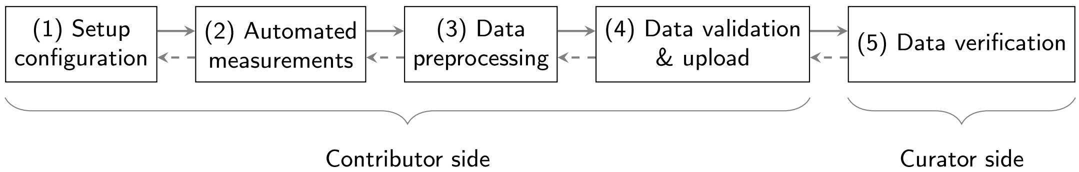

Fig. 4. The simplified process of conducting crowdsourced measurements. Feedback loops between all steps allow for iterative improvements and error handling.

only if they provide such a ground truth. The labelling requirements, which include device type, vendor, model, device state, and measurement environment, enable the data to be applied flexibly to other research contexts. This flexibility could potentially lead to the identification of previously unknown dependencies in device behaviour. Consistent and comprehensive labelling is supported by the automated measurement system provided to contributors, as described in the following section. Apart from ground truth and labelling, we set no other requirements pertaining to device state, environment, measurement hardware, shielding boxes etc. to facilitate easy contributions and to allow the data to be used for a wide variety of research projects.

A successful crowdsourcing methodology must demonstrate not only strategic expertise (incentive systems, governance, transparency) but also technical excellence (data quality, anonymization, storage management). Scientific analyses on CNMs already exist, such as the study by Hirth et al. [17], which compare CNMs with traditional measurement techniques, identify potential pitfalls and limitations, and outline best practices for developing a robust crowdsourcing methodology. While CNMs typically refers to measurements based on packet data, in this context we focus specifically on the controlled collection of PR data.

In Fig. 4, the process of conducting automated measurements is illustrated. We envision the process as follows: The setup configuration (1) involves creating a configuration file that defines all relevant parameters for the subsequent steps. Based on this configuration, automated measurements (2) are performed, during which data is collected automatically or semi-automatically. These initial steps are described in more detail in Section 6.1.

In step (3), data preprocessing, the collected data undergoes processing operations such as anonymization to ensure compliance with data protection requirements. Following this, data validation & upload (4) includes automated checks to verify the integrity and quality of the data before it is uploaded. Finally, in step (5) data verification, the data must be manually reviewed and approved before it is permanently written into the database. These latter stages are partially outlined as challenges and solutions in Section 6.2. Feedback loops between all steps allow for iterative improvements, or error handling and ensure an overall robust and reliable workflow. The key to effective crowdsourcing practice is the careful design of measurement automation, which will be discussed in the following section.

6.1. Automating measurements

The biggest challenge in growing the size of the data set, apart from device procurement, is the manual and time-intensive nature of the measurements. Therefore, we see automation of the measurement process as a central challenge in facilitating contribution to the data set. For Android devices, ADB can be used to gather information about a device and control its state. This way, the measurement script can automatically extract device data (e.g. vendor, model and OS version) and change the device state between measurements without user intervention. The user only needs to enable ADB, plug the device into the measurement computer and start the measurement script.

Our automated Wi-Fi capturing system is based on a laptop running Ubuntu Linux which is connected to one or more USB Wi-Fi adapters (one per channel that is monitored). A custom Python script controls the measurement process using the Linux tools `tcpdump`, `ip`, `iw-config`, `iw`, `mergecap` and `tshark`. It sets up the Wi-Fi adapters, manages the device state and starts and stops `tcpdump` processes to capture the probe requests emitted by the device under test. It also performs some postprocessing (e.g. RSSI filtering) and generates a `YAML` file containing metadata about the measurement. For devices that support ADB, the script requires no user intervention after configuring and starting the script. For devices that do not support ADB (e.g. iOS devices), the script interactively guides the user through the measurement process. In the beginning, it asks the user to enter the device data that is otherwise extracted via ADB (i.e. manufacturer, model, OS version etc.). It then sets up the Wi-Fi adapters, begins the measurement and prompts the user to manually change the device's state when necessary. After the user confirms that the required changes have been made, the measurement continues. Wi-Fi adapter management, postprocessing and metadata generation are the same whether ADB is used or not.

Before starting the script, a `YAML` file is used to configure the required information. It sets the interface names of the Wi-Fi adapters, the channels to be monitored and a list of measurements that are performed sequentially. Each measurement run is defined by a device state, a duration in seconds and some metadata for labelling. The device state is defined by a number of ADB commands that are executed before beginning the measurement. When started, the script first enables monitor mode on the respective Wi-Fi interfaces and sets them to the correct channels. Then, it sequentially executes the measurements defined in the config `YAML`. For each measurement, the respective ADB commands are executed or the user is asked to enter the required state manually. Then, several `tcpdump` processes are spawned to capture probe requests on all interfaces. After waiting for the defined number of seconds, these processes are terminated and the resulting `PCAP` files are merged into a single file using the `mergecap` tool. Then, optional filtering by RSSI is performed using the `tshark` tool.

To make sure that only frames originating from the device under test remain after RSSI filtering, the device is placed immediately next to the antennas of the measurement setup, and all other Wi-Fi devices are placed at least some metres away. Thus, frames are received with a very high signal strength if and only if they originate from the device under test. To further improve this setup, the entire measuring system can also be placed inside a shielding box, if one is available.

After finishing the measurements, the script generates a metadata `YAML` file describing the captured data. It contains information about the measurement runs (respective states and durations), the device (vendor, model, OS version) and the measurement setup (adapters and channels, environment, distance to other devices etc.). Device data is either obtained through ADB or otherwise entered manually during the measurement process. The other metadata is taken from the configuration of the measurement script. With this metadata, the PCAPs can directly contribute to the data set via our inclusion script.

The use of ADB requires the device to be connected to a computer, thus being charged while the measurement takes place. We are not aware of any findings indicating that devices behave differently while charging with respect to Wi-Fi probing, but we did not verify this assumption. Therefore, we still run the ADB command `dumpsys battery unplug` before starting the measurement. It simulates unplugging the device without actually cutting the ADB connection and causes the system to behave as if it were not plugged in [18].

6.2. Crowdsourcing challenges and solutions

Crowdsourcing allows us to grow the data set much faster and larger than by only performing measurements ourselves. However, crowdsourcing also requires the definition of clear requirements for contributions, effective quality control measures and thorough anonymization of submitted data. In the first step, we identified the major challenges posed by a crowdsourcing approach and derived specific requirements to address them. One challenge is ensuring the consistency and quality of the contributed data, along with proper documentation and privacy-compliant data handling. Another challenge is the trade-off between data consistency and standardization on the one hand and the benefits of device and environment variation. The third challenge lies in ensuring the sustained engagement of contributors by effectively motivating them to generate and submit measurements, while simultaneously securing the long-term feasibility and sustainability of the project.

To ensure quality control over contributed data, we provide scripts as well as documentation on how to set up the measuring system and how to use the scripts. A contributor should be able to perform measurements using these provisions without additional assistance. The guidelines include notions on how a measurement is performed with which hardware options, how long measurements should last at minimum and how the data will have to be labelled to be accepted. For the measurement environment, a list of recommended Wi-Fi adapters and environmental parameters are provided, i.e. notions about the usage of shielding boxes, RSSI-thresholds or cross-channel interference in 802.11n. That way, we aim to assure consistent measurement profiles and a labelling state we would have hoped to observe, when looking for other PR data sets. Once measurements have been conducted, a postprocessing step is required before the data can be uploaded using a method detailed on the data set website. Data uploaded using that method will be checked for adequate labelling and other compliance factors, such as successful anonymization. The acceptance criteria will be communicated clearly and transparently for potential contributors to abide by.

A provided automation script supports measurements using the ADB interface. Additionally, the script aims to assist measurements with non-ADB-compatible devices by guiding the contributor through device labelling and the measurements itself, as described above. A sample configuration file with the measurement protocol described in Section 4.2 is provided. For postprocessing, a script is provided that transforms the measured data into the correct format for import into the data set. It also anonymizes non-randomized MAC addresses and SSIDs contained in the measurements to ensure privacy. For the anonymization we use `argon2id` with a random salt per device, where an SSID is anonymized to an expression starting with "SSID-<hash>".

According to the fact that devices behave differently in sending PRs depending on their direct radio environment and user level-configured networks, it is necessary to allow some degrees of freedom in the measurement setups. The resultant differences in radio environments and device configurations add valuable diversity to the data set. To ensure reliability and reproducibility, the documentation process should be automated and should follow a defined adaptable scheme that captures essential metadata, enabling easy repeatability of measurements under varying conditions. Alongside plausibility checks prior to upload, a scoring or weighting system assesses the credibility of each data source, promoting contributor confidence while safeguarding against malicious or unreliable submissions.

To address the third challenge, the process of recording measurements and submitting them to the project should be made as easy as possible. To achieve that, we do not require contributors to use specific or specialized hardware. The only requirements are a laptop or computer running Linux and one or several USB Wi-Fi adapter that support monitor mode operation. A how-to guide for contributing measurements will be provided with the data set. The measurement automation script described in the previous section also helps to minimize the effort required, both for us and possible contributors. We have already reached out to potential participants from our research network during the work on this paper. This has led to the addition of a number of devices to the data set as seen in Section 4. Additionally, we intend to use conferences and other social gatherings as well as science communication channels to attract contributors.

The feasibility of crowdsourcing for data collection has been demonstrated in a number of different contexts within the research area of computer networks. For example, Lohan et al. [19] use a crowdsourcing methodology to compile a data set for indoor Wi-Fi based positioning. They provide an Android application that contributors install on their phones prior to moving around the building and collecting signal strength data at their respective location. Hirth et al. [17] compiled a detailed study on crowdsourcing-based data collection for Internet-wide CNM as an alternative to passive observation or locally administered experiments. They give an overview of (dis)advantages of CNM, its use cases and best practices. Their crowdsourcing approach relies on paid crowdsourcing platforms (e.g. Amazon Mechanical Turk) to recruit contributors for a fee. The Wigle.net platform [20] is the result of a large and ongoing crowdsourcing effort to compile a global database of Wi-Fi networks with their respective SSID, location and security settings. As of 2025, the project has been running for 23 years, contains more than 25 billion Wi-Fi observations and covers most of the globe. This shows that it is indeed possible to collect large amounts of data through crowdsourcing, and that this type of data sets can also grow sustainably for many years.

7. Discussion and future work

Our presented work does have limitations regarding its validity on broader data than the data that has been used to conduct the described examination above. An in depth discussion on the limitations of both our available data sets as well as the performance of templating and our identification algorithm are presented in this section. We will then point out future research directions and possible countermeasures against the presented approach. A discussion on ethical considerations is following after that.

7.1. Data set limitations

The evaluation of any DI approach is highly dependent on the quality of the data set(s) it is evaluated with. Currently, our newly published data set is still rather small in size, covering 15 devices in three different states, introducing several data selection biases, as mentioned in Section 4.2. Comparing the capturing modes between different data sets, a similar problem is apparent. The UJI, Cagliari and Bonn have one common measurement mode, which only a small part of the Cagliari and Bonn data set was measured in. These data sets were compiled with different goals in mind, thus diverging in experimental design and measurement setup. This limits the generalizability of our findings regarding the impact of MAC randomization on the PRs and their generated templates. Yet, to the best of our knowledge, there is no other high-quality data set of probe requests with both MAC randomization and a ground truth.

Since our presented algorithm uses a knowledge base that cannot generalize towards PRs that it was not built upon, the limited variety of devices and capturing modes would make our algorithm volatile when trying to use it on never seen data. On the one hand, any PR that fits an entry in the knowledge base will be identified as a corresponding device, which given the market coverage of our currently available data set can succeed in correctly identifying a sending device depending on the measurement location. On the other hand, any PR matching an entry but stemming from a device not yet featured in our knowledge base will be incorrectly attributed to a different device. Furthermore, any PR not matching any entry simply cannot be attributed. Given the discussions in Sections 5.2, 5.3, and 5.4 we argue, that without deeper understanding and a vastly more diverse data foundation, any attempt at generalizing PR attribution to specific device models purely based on IE content is a difficult problem to solve. For example, based on our available data, there is no telling, how the PRs of a Sony phone will be constructed, or any Google phone released in the next few years. However, using our templating approach for data set filtering and thus structurally splitting incoming data can make the data more closely fit towards specifically trained classifiers for a given template. We hypothesize that using templating can significantly improve the performance of trained classifiers, though we do not show this hypothesis in this work. Changing software or hardware on a phone, firmware or other software updates also emphasize the need for a system for continuous knowledge capturing, as newer releases can have an impact on the observable data. At the current state of research, a study on behavioural differences of templates on operating system or firmware updates is not possible. However, given a substantial knowledge base relative to the observed data, we demonstrated that an analytical approach can yield results not lacking behind the presented research peers.

Our goal is to mitigate these limitations and better document the data set. A growing data set will introduce an increasing identification accuracy and versatile analysis capabilities. Particularly, an extensively documented heterogeneous data set and more metadata will allow for more detailed knowledge base, for example for the identification of subtle patterns using templates. We aim to grow our data set into a comprehensive collection of recordings from a broad spectrum of devices in different states, environments and using different measurement setups. To this end, we will encourage contributions by the research community as well as contribute under-represented devices ourselves. This can help assist in many different research directions where there is not yet a consistent base of data sets sufficient in size. This will finally introduce an excellent basis for future work. As measurements of numerous consumer devices can hardly be carried out by a single party, we introduced our intended crowdsourcing-based approach in Section 6. Increasing the size and variation in devices contained in the data set will improve the generalizability of our findings. The growth of the data set must be managed and maintained with regards to data quality, measurement setups and further developments in devices and technologies.

7.2. Templating performance

The evaluation of template characteristics has revealed that templates are highly diverse (especially in modern data sets), many devices have unstable templates, though rarely more than two different ones, and contribute to a significant compression of all of the considered data sets. For the unlabelled data sets, the magnitude of outlier templates is surprising. However, this may be caused by a large number of observed devices actually belonging to a small family of very commonly used device models. For example, it is conceivable that all 100 devices sharing a template in the UJI data set are actually different specimens of the same phone model (a very common model from a popular manufacturer) or even the exact same phone, which employs MAC randomization without setting the *global bit* as described in Section 4.1. In the labelled data sets, which achieve much more even template distribution, almost all phone models are only included once. This shows a certain imbalance in data sets consisting of PRs captured in the wild, where the common marked distribution of Wi-Fi devices owned by the public may affect transferability of the findings to other data sets. Having a larger crowdsourced data set from contributors from different regions can allow to mitigate certain biases, when the crowdsourcing is managed correctly.

The bias of unlabelled data sets also demonstrates a potential weakness of our DI approach, and of all PR-based fingerprinting approaches: Two devices of the same manufacturer and model, especially if using the same operating system and firmware version (e.g. the most recent version), will only differ in configuration and state. While configuration and state do have an impact on PRs, it is likely that they will generate very similar PRs, resulting in similar or identical templates and FPs. Our evaluation and previous work have demonstrated that probing behaviour is influenced by vendor, model, operating system, chipset and firmware version, configuration, state and environment. However, it remains unclear which of these factors results in which differences in templates and FPs. A detailed analysis of the influence of each of these factors for specific phone models would be required, which we leave for future work.

We evaluated our DI approach with the available data sets to attribute Wi-Fi frames to individual devices without relying on MAC addresses, which we assume to be effectively randomized. The evaluation shows that our proposed PR templating-based DI approach achieves a drastic performance improvement over naive fingerprinting for that particular observation of PRs. These results highlight the potential value that a large knowledge base could offer to DI systems. It not only improves the DI accuracy but also provides explainable and trustworthy results by design, something that machine learning methods often struggle to deliver. The application of expert knowledge for PR templates promises a meaningful performance improvement. Taking the initially presented use case into account, our goal was to identify and separate packets of an individual device within an anonymity group of limited size observed in spatially and temporally limited scenarios such as a burglary. Assuming an extensive knowledge base is established and available, we expect that PR templating assisted DI is a competitive and reliable solution to the problem, delivering trustworthy and explainable results.

In comparison with the results of other research e.g. on the Cagliari data set, we need to account for the difference in use cases. Pintor et al. [8] use clustering methods and compare their amount of clusters with the actual device count. In our case, this is equivalent to counting the number of optimized template/FP groups, which yields 20 groups for 22 actual devices. Though the goal of our approach is unique separation of devices instead of device counting, our approach still works reasonably well for counting, at least on our test data sets. However, we also need to consider additional features (e.g. sequence numbers and signal strength measures) for device counting in real-world scenarios, as multiple devices of the same type may potentially be present.

Though we have shown that templating-based DI performs reasonably on all available data sets published past 2022, we are confident that templating has merits outside of the examined data sets. Since our approach is analytical and deterministic in nature with comprehensible results, there is no effect like overfitting to be had for a specific data set. Instead, we expect our approach to vary in performance for more homogeneous data sets, i.e. an all iPhone data set or for older data sets, as discussed with Sapienza. However, when relying on an already established knowledge base to perform DI using our approach, that knowledge base must first be established using diverse PR measurements. The more different devices were used to fill the knowledge base, the less likely it is to encounter a never seen before template. On the other hand, the more different devices were used to fill the knowledge base, the more likely will templates be shared between different devices that have been seen. We suspect that it is highly unlikely, if not impossible, for any one device to have a truly unique template or FP when set against all other Wi-Fi devices on the planet. DI during a simultaneous observation of Wi-Fi traffic of all devices on the planet is not a use case that we try to solve. Our approach and likewise any other fingerprinting approach can only work with a limited amount of simultaneously observed devices, limiting the anonymity group the target device must be identified within. For any given previously unseen anonymity group without a given ground truth our approach may fail to correctly identify a target device. This happens if in that group other observed devices share templates or FPs and the knowledge base cannot solve the overlap.

7.3. Future work

In addition to the sheer size of the data set, a number of technical nuances were either not fully considered or not considered at all. For example, it remains unclear how the proposed approaches would translate to the 5 GHz band, as this may involve different propagation characteristics and interference patterns. Additionally, the dependencies on operating systems or vendor chipsets were not fully explored. In particular, potential changes caused by firmware updates may have an impact on device behaviour. Precisely understanding a device's behaviour in all circumstances requires full reverse-engineering of the device's Wi-Fi stack, which is unrealistic. In addition to different operating system and firmware versions, it is also not known how the devices might behave in a different environment. Some studies show that a higher number of PRs may be transmitted when beacon frames from known Wi-Fi networks are received [21]. This could result in unrealistic behaviour if using a shielding box (as done in some of our measurements). In busy environments, the CSMA/CA mechanism can also lead to unpredictable variations in PR timing, which may throw off fingerprinting algorithms that rely on periodicity analysis. Controlling for these factors would require a shielded room or box and a number of fully controlled, precisely monitored devices that create a realistic, but reproducible radio environment. Such a setup is very complicated and thus outside the scope of this work.

Since we cannot control all of these factors, we instead aim to document them as thoroughly as possible. As described in Section 6.1, with each PCAP file we include a detailed description in `YAML` format. This description includes information about the measurement setup (environment, measurement devices etc.), the target device (OS version, model name etc.) and the target's state (power saver, Wi-Fi connection etc.). The information can be used by future researchers to exclude measurements that are likely to contain significant bias in their specific application scenario.

All these investigations may bolster our approach and other fingerprinting approaches and thus we consider them for future research. For these and other future research directions, we are certain that a well managed large crowdsourced data set will prove invaluable for us and other researchers interested in Wi-Fi PRs and device behaviour. By leveraging the automation of the measurements presented here and the growing network of collaborators we have established, we are confident in our ability to overcome the challenges involved and make a significant contribution to advancing this area of research. Finally, a detailed examination of the practical performance of these device-identifying data collections can then be conducted. In particular, we are interested in their effectiveness in actual device re-identification scenarios, such as in the context of criminal investigations.

7.4. Countermeasures

Based on our observations, PR templates are a useful tool for re-identifying devices due to their diversity between different devices and, at the same time, relative stability per device. If full mitigation of PR templating is required, it may be possible to forego PR entirely, instead relying on passive network discovery based on Access Point (AP) beacons. This mitigation, however, is inconceivable because PR are important for performance, roaming, and compatibility. Instead, privacy improvements are achieved through randomization and more restrictive policies. It requires considerable modifications of clients and would most likely lead to higher battery drain, longer time-to-connect and thus worsen usability of the system. Standard-conforming countermeasures that keep PRs can work towards mitigating either template diversity or template stability to prevent effective fingerprinting. While manufacturers have some flexibility in PR structure and content, IEs carry information that is essential for connection establishment and therefore cannot be altered arbitrarily.

To mitigate template diversity, manufacturers could try to remove as many IEs as possible or they could collaborate to create a standard in order to achieve a common representation. Such a standard should be as compact as possible without sacrificing functionality and must include all important IEs, even if a device does not support the corresponding feature. This would make PRs all look the same, reducing the ability of templates and FPs to differentiate between different devices. Including too many optional IEs in the standard pool inflates the size of control frames and reduces efficiency. On the other hand, leaving out certain IEs or restricting the available IE pool may limit current or future functionality. Template stability per device could be reduced by introducing additional, random information elements and values and/or by randomizing the order of the included information elements. Similar randomization techniques are also recommended and applied against fingerprinting in TLS (c.f. GREASE/RFC8701 [22]). Such randomization results in a larger variation in the templates and FPs of a single device, preventing the identification of one device's PR based on a single template or FP. The additional elements must be sufficiently realistic that they cannot be separated from the "real" IE content, but also not interfere with the function of IEs, i.e. they must not advertise features that the device does not support. While this type of obfuscation can be effective, it inflates the size of control frames, resulting in higher power consumption, increased strain on the wireless medium and a general loss off efficiency.

While these strategies are effective against the fingerprinting discussed in this paper, they can still be overcome by more advanced fingerprinting methods. Against improved fingerprinting methods (not covered in this paper), it may be necessary to randomize even the timing of when PRs are transmitted, which currently depends on vendor firmware (c.f. Fig. 3 in Rusca et al. [23]). Nevertheless, it seems that some countermeasures are already being taken by individual manufacturers. Gu et al. [24] observes that Apple generates random noise according to variations in certain PR IE parameters on the iPhone X. Overall, this type of countermeasure has a greater impact on established fingerprinting procedures than on our templating-based approach, depending on how exactly the countermeasure is implemented.

7.5. Ethical consideration

The templating based DI generally allows privacy-aware analysis of end devices without deanonymization of individuals. From a privacy perspective, our fingerprinting method is no different from previous PR fingerprinting approaches. Our approach does not allow random MAC addresses to be resolved to the original hardware address in order to identify a specific end device. However, it is still possible to track an individual device by correlating PR FPs with spatial radio information and sequence numbers, e.g. demonstrated within the work of He et al. [25]. But, long-term physical tracking of mobile devices remains difficult in practice due to several factors: PRs are sent irregularly and less frequently in different states of modern devices, signal strength can fluctuate due to environmental conditions, and device FPs may change after software updates or configuration changes. Additionally, reliable triangulation requires dense sensor coverage, which is rarely available outside controlled environments. Furthermore, the SSID as another sensitive value is not evaluated or processed by our DI approach. Without using SSIDs wildcards, PRs will be sent with partly confidential information, as some analyses have already shown. E.g., Barbera et al. [26] demonstrate how SSIDs from PRs can be misused for social analysis and profiling. This allows for individual user identification and tracking, inferring relationships between users in a crowd, or inferring user languages and in individual cases even user names. Today, most devices use SSID wildcard probe requests to enhance privacy by avoiding the disclosure of previously known networks. Although SSIDs are transmitted in plaintext, they are not evaluated by our approach; instead they are anonymized in the data set.

In addition to the ethical considerations concerning the fingerprinting methodology, there are also important implications when creating large data sets of probe requests from end devices. A very relevant threat exists in group profiling, such as the recognition, identification and tracking of groups. In smaller groups or with rare devices and firmware versions, the risk of creeping deanonymization increases. With the help of additional context data, it may even be possible to re-identify individuals at a later time. Thus, large, very precise data sets may allow very individual classifications. Furthermore, a large-scale data set containing fine-grained technical characteristics of vendors, devices, and models can enable indirect inferences about users' socioeconomic status, geographic origin, or age groups, based on correlations with specific device types or manufacturers. E.g., when certain device categories (such as low-cost Android models) are overrepresented, this can lead to biased interpretations, reinforcing harmful stereotypes. In worst-case profiling scenarios, this could result in discriminatory practices, such as access restrictions, differential treatment in public spaces, or targeted surveillance of marginalized groups. The technical foundation for these sorts of infractions will likely stay as long as modern Wi-Fi devices are in use, necessitating legislative intervention as soon as technologies using the presented methods see widespread use.

8. Conclusion

In this paper, we propose a novel Wi-Fi PR templating-based device identification approach. We describe the construction and usage of Wi-Fi PR templates, how to use them for creating a knowledge base and how to evaluate their quality. We apply this templating approach to four PR data sets in order to measure the amount of templates generated by single devices, the amount of devices that share templates and the compression rate that the filtering using a template has on the whole data set as well as its accuracy in device identification.

We describe the evolution of PR templates by applying it to Wi-Fi PR data sets captured more than 10 years apart. We find that the amount of IEs contributing to the generated templates increased and thus a more diverse set of templates helps to differentiate modern devices notably better. A series of experiments is conducted to evaluate the impact of our PR templating assisted device identification approach. We showed that most devices share all of their observable information within the first minute of observation. The DI approach is applied to four data sets of PRs captured in different scenarios in order to separate individual devices from all other devices solely based on the content of their sent PRs. We found our approach to perfectly separate 75 % to 85 % of all devices for data sets captured past 2022 that were used for the experiments. Since the knowledge base does only comprise data measured from a limited set of devices, our findings do not reflect the approach's performance on a larger data set or its capability of generalization on unseen data. Still, when properly prepared, these information could be used by law enforcement to re-identify devices that have been observed at crime scenes in order to strengthen home security or threaten the digital sovereignty of users of Wi-Fi devices, when used with malicious intent.

Additionally, we found that PR templates can prove as valuable assets towards device counting, as successfully separating devices allows to also estimate the amount of devices present at a given time. Unintentionally, we also found that our templating method can assist in PR data set cleansing, preventing mislabelling measurements for devices that are meant to be separated by signal strength or randomized MAC addresses.

In summary, to answer whether a single Wi-Fi device can be separated from all other nearby Wi-Fi devices solely based on PRs the set of all observed devices needs to be assessed. In a data set, where devices are sufficiently heterogeneous in terms of vendor, model, software and settings, it is not only possible to separate the device, but also to link most PR sent by a device correctly back to it.

In order to lay the groundwork for future research, we introduce a crowdsourcing-based process to generate a data set that will serve as a basis for various research questions. For this we propose an abstract process encompassing automated measurements while also addressing challenges and solutions related to data validation and verification, without overlooking ethical considerations.

Despite the meaningful results obtained, we discuss strengths and weaknesses regarding PR data sets as well as limitations and possible countermeasures. For instance, the amount of devices and variations in the data sets we could use was relatively small, as well as previous data sets were often not sufficiently documented. Other technical nuances and deviations regarding the measurement setup, target device characteristics and target's state information need to be considered, documented and analysed. Still, our approach will not be able to generalize to identify devices of which there are no entries in the knowledge base. We aim to gradually achieve this with our current data set and the future crowdsourcing procedure. Nevertheless, we must also take into account that there could be countermeasures that might undermine PR fingerprinting in general, and potentially our templating-based device identification approach as well, which we reveal in our discussion.

Finally, in future we will dedicate efforts to achieve a scalable crowdsourcing-based approach for the generation of Wi-Fi PR measurement data, which will assist our research to come as well as may assist other researchers in their investigations. Following on from our existing presentation, we will provide guidelines on how to measure, label and submit data, which will be accepted into the data set, if acceptance criteria are fulfilled.

Future research will aim to address the identified limitations and to further investigate the discussed research questions as well as the integration of our templating approach into existing real-word scenarios. In conclusion, this work makes a contribution to Wi-Fi PR fingerprinting, offering new perspectives and valuable insights that pave the way for further advancements in the field.

CRediT authorship contribution statement

Daniel Vogel: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Felix Viola:** Writing – original draft, Validation, Software, Methodology, Investigation, Data curation. **Nicholas Malte Kreimeyer:** Writing – original draft, Validation, Software, Investigation, Data curation. **Daniel Bücheler:** Writing – review & editing, Software, Project administration. **Sebastian Böhm:** Writing – review & editing, Project administration, Funding acquisition. **Michael Meier:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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```
knowledge_base: {
    "tags;supra;extrates;DS_channel;221_0050f2_08": {   // template bin
        "supra": {                                      // separator for fingerprint
            "supra:82848b96": "Google Pixel 9",
            "supra:02040b16": "Google Pixel 3A",
        },
        None: "Google Pixel 9",                         // default for this template bin
    },
    "tags;supra;extrates;221_0050f2_08;DS_channel": {
        "221_0050f2_08": {
            "221_0050f2_08:001100": "Xiaomi Redmi 9",
            "221_0050f2_08:001500": "Motorola Moto G23",
        }
    },
    "tags;supra;extrates": {None: "Generic Template"},
    ...
}
```

Fig. 5. Excerpt from the build Knowledge Base used in this paper to create the given results. It is structured as a list of templates, that contain template rules to differentiate between devices with the same template; applying this template results in a minimal fingerprinting, separating these devices. A default `bin_name` can be given with the `None` keyword.

Appendix A. Pseudocode

This section goes into more detail about the design of the knowledge base and the used separation algorithm. First, the knowledge base will be discussed and together with an excerpt in Fig. 5. After that Algorithm 1 is discussed, that uses the knowledge base.

The general structure of the knowledge base is composed of a list of templating groups that further contain templating rules, that – when applied – result in minimal fingerprints, separating devices within these templating rules. In the given excerpt in Fig. 5, the first entry is the template `tags;supra;extrates;DS_channel;221_0050f2_08`, that gets exclusively created by the Google Pixel 3 A and 9 in our data sets. These two devices can be separated based on the IE supported rates (`supra`), being the templating rule used here. When calculating the fingerprint using this rule, it can be matched against the given list of known fingerprints. E.g. in case the resulting fingerprint is `supra:02040b16`, this would match for the Google Pixel 3 A. However, in case none of these templating rules match, a default name can be provided for the templating group. This is used when either all templating rules "filter out" all other device and only one device is left, or in case devices cannot be separated and grouped instead — e.g. in the `Generic Template`.

To give an overview of the algorithm, first all packets are grouped based on the calculated template into bins. In each bin, the knowledge base is applied to see if templating rules are present to separate devices further, into their own bins. Once all bins contain individual devices or are groups of devices not separable, the knowledge base supplies a label for each bin. More detail into these steps is given in the following.

If a device is individual in its templating group, no further separation is needed. The name of the device can be looked up in the knowledge base. However, as other devices can still share the same template but might not be present, the knowledge base must be further checked. By applying templating rules and checking if the resulting fingerprint is known, the correct device name can be found. If the device is alone in its templating group, a default name is given. If multiple devices are in-differentiable, a name is given, under which these devices are grouped.

Finding the minimal separating fingerprint for a set of devices, in case multiple devices generate the same template, is done by searching for a fixed set of IEs that perfectly separate these devices. This is done by iterating over a bitmap of present IEs in the template, calculating a shorter fingerprint with only this selected set of IEs. If no such separation can be found, it is not possible to further separate these devices; so the knowledge base supplies a default name for the device group. But if a separator is found, the same steps are applied on every device, as described in the above individual device case.

Finally, if the knowledge base does not supply any bin name, it is marked as `unknown`. By analysing the device, its information could be added to the knowledge base to be identified in future executions.

Appendix B. Observation time plots

To understand, how long a device needs to be observed in order to observe all observable data that this device is transmitting, we examined the timings of when new information was observed relative to the observation length as well as other devices in the same measurement modes AR (cf. Section 3), PS and S (see Figs. 6 and 7). An in-depth discussion on the observations is given in Section 5.3.

Appendix C. Confusion matrices

The following matrices show the output from the applied algorithm, mapping probe requests from devices on the input side (left column) to the named bins on the right. Given is the number of packets assigned to the according bin. The cells are coloured green if the mapping is correct, orange if the target bin is a group for multiple devices (because no further separation is possible) and red, if the mapping is wrong.

Results from our data set `Bonn` in modes AR, PS and S can be found in Table 12, Table 13 and Table 14. For `cagliari`, the results are in Table 15, Table 16 and Table 17 respectively. Finally, Table 18, Table 19 and Table 20 show the results when combining both data sets.

Appendix D. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.comcom.2025.108367.

Data availability

A DOI to the data set and further instructions is mentioned inside the paper.

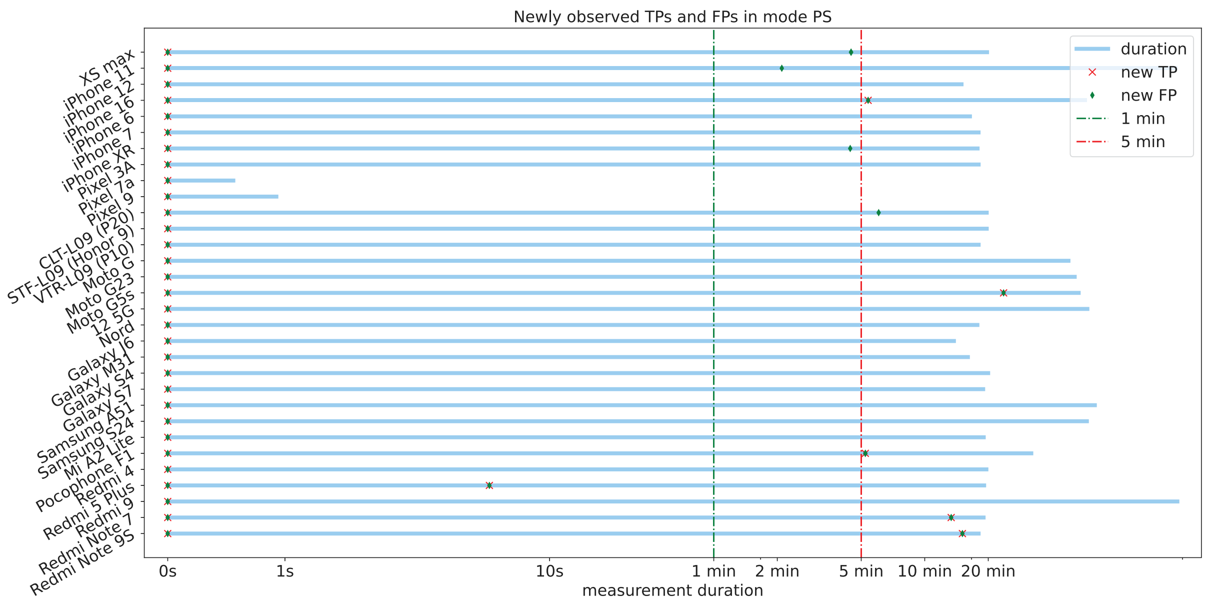

Fig. 6. Timings of newly observed templates (TP) and fingerprints (FP) relative to the overall measurement duration for various different devices. Only four devices showed new information past five minutes of observation time.

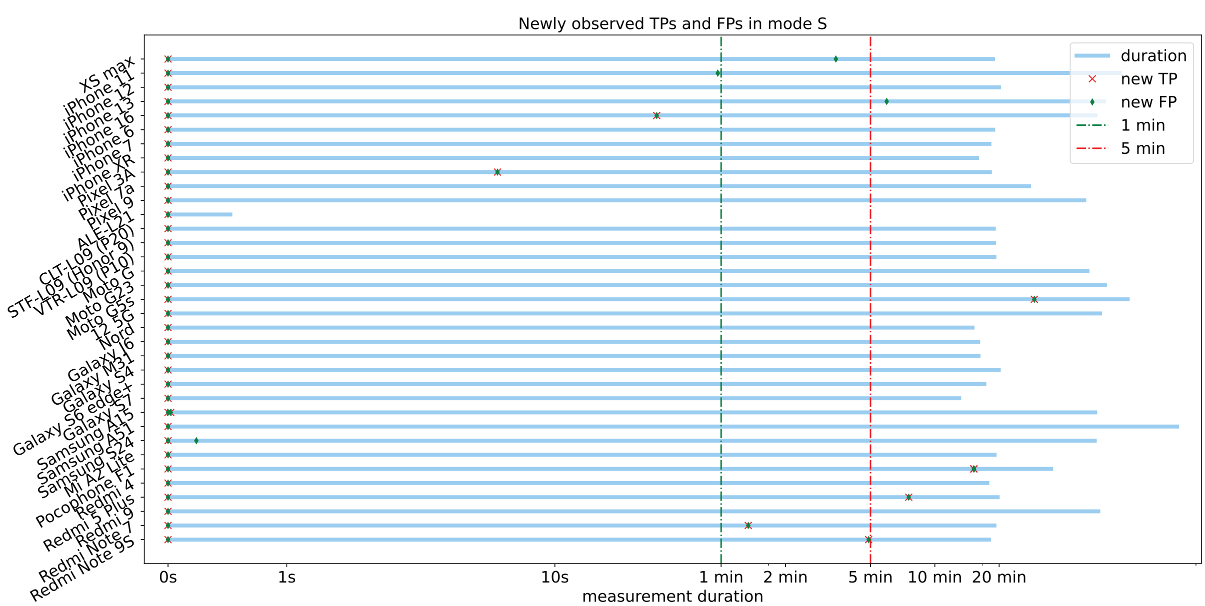

Fig. 7. Timings of newly observed templates (TP) and fingerprints (FP) relative to the overall measurement duration for various different devices. Only four devices showed new information past five minutes of observation time.

Table 12
Device Classification Confusion Matrix - Datasets: bonn in mode: AR.

	Apple iPhone 13	Apple iPhone 16	Google Pixel 7a	Google Pixel 9	Motorola Moto G	Motorola Moto G23	OnePlus 12 5G	Samsung Galaxy A15	Samsung Galaxy S24	Xiaomi Pocophone F1
Apple iPhone 13	238	0	0	0	0	0	0	0	0	0
Apple iPhone 16	0	259	0	0	0	0	0	0	0	0
Google Pixel 7a	0	0	1241	0	0	0	0	0	0	0
Google Pixel 9	0	0	0	380	0	0	0	0	0	0
Motorola Moto G	0	0	0	0	988	0	0	0	0	0
Motorola Moto G23	0	0	0	0	0	197	0	0	0	0
OnePlus 12 5G	0	0	0	0	0	0	1009	0	0	0
Samsung Galaxy A15	0	0	0	0	0	0	0	354	0	0
Samsung Galaxy S24	0	0	0	0	0	0	0	0	1	0
Xiaomi Pocophone F1	0	0	0	0	0	0	0	0	0	1294

Table 13
Device Classification Confusion Matrix - Datasets: bonn in mode: PS.

	Apple iPhone 11	Apple iPhone 16	Google Pixel 7a	Google Pixel 9	Motorola Moto G	Motorola Moto G23	OnePlus 12 5G	Samsung Galaxy A51	Samsung Galaxy S24	Xiaomi Pocophone F1	Generic Template	Moto G5s OR Redmi 5 Plus
Apple iPhone 11	309	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 16	0	285	0	0	0	0	0	0	0	0	0	0
Google Pixel 7a	0	0	18	0	0	0	0	0	0	0	0	0
Google Pixel 9	0	0	0	6	0	0	0	0	0	0	0	0
Motorola Moto G	0	0	0	0	7	0	0	0	0	0	0	0
Motorola Moto G23	0	0	0	0	0	93	0	0	0	0	0	0
Motorola Moto G5s	0	0	0	0	0	0	0	0	0	0	129	33
OnePlus 12 5G	0	0	0	0	0	0	180	0	0	0	0	0
Samsung Galaxy A51	0	0	0	0	0	0	0	144	0	0	0	0
Samsung Galaxy S24	0	0	0	0	0	0	0	0	78	0	0	0
Xiaomi Pocophone F1	0	0	0	0	0	0	0	0	0	1001	0	0

Table 14
Device Classification Confusion Matrix - Datasets: bonn in mode: S.

	Apple iPhone 11	Apple iPhone 12	Apple iPhone 16	Google Pixel 7a	Google Pixel 9	Motorola Moto G	Motorola Moto G23	OnePlus 12 5G	Samsung Galaxy A15	Samsung Galaxy A51	Samsung Galaxy S24	Xiaomi Pocophone F1	Generic Template	Moto G5s OR Redmi 5 Plus
Apple iPhone 11	400	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 13	0	171	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 16	0	0	376	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 7a	0	0	0	42	0	0	0	0	0	0	0	0	0	0
Google Pixel 9	0	0	0	0	36	0	0	0	0	0	0	0	0	0
Motorola Moto G	0	0	0	0	0	12	0	0	0	0	0	0	0	0
Motorola Moto G23	0	0	0	0	0	0	104	0	0	0	0	0	0	0
Motorola Moto G5s	0	0	0	0	0	0	0	0	0	0	0	0	155	49
OnePlus 12 5G	0	0	0	0	0	0	0	183	0	0	0	0	0	0
Samsung Galaxy A15	0	0	0	0	0	0	0	0	452	0	0	0	0	0
Samsung Galaxy A51	0	0	0	0	0	0	0	0	0	274	0	0	0	0
Samsung Galaxy S24	0	0	0	0	0	0	0	0	0	0	49	0	0	0
Xiaomi Pocophone F1	0	0	0	0	0	0	0	0	0	0	0	1009	0	0

Table 15
Device Classification Confusion Matrix - Datasets: cagliari in mode: AR.

	Apple XS max	Apple iPhone 12	Apple iPhone 7	Apple iPhone XR	Google Pixel 3A	Huawei ALE-L21	Huawei CLT-L09 (P20)	Huawei STF-L09 (Honor 9)	Huawei VTR-L09 (P10)	OnePlus Nord	Samsung Galaxy J6	Samsung Galaxy M31	Samsung Galaxy S4	Samsung Galaxy S6 edge+	Samsung Galaxy S7	Xiaomi Redmi 4	Xiaomi Redmi 5 Plus	Xiaomi Redmi Note 7	Xiaomi Redmi Note 9S	Generic Template	Xiaomi Mi A2 Lite OR Redmi 5 Plus
Apple XS max	179	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 12	0	59	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 7	0	0	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone XR	0	0	0	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 3A	0	0	0	0	123	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei ALE-L21	0	0	0	0	0	1504	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei CLT-L09 (P20)	0	0	0	0	0	0	246	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei STF-L09 (Honor 9)	0	0	0	0	0	0	0	317	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei VTR-L09 (P10)	0	0	0	0	0	0	0	0	119	0	0	0	0	0	0	0	0	0	0	0	0
OnePlus Nord	0	0	0	0	0	0	0	0	0	143	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy J6	0	0	0	0	0	0	0	0	0	0	240	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy M31	0	0	0	0	0	0	0	0	0	0	0	113	0	0	0	0	0	0	0	0	0
Samsung Galaxy S4	0	0	0	0	0	0	0	0	0	0	0	0	3403	0	0	0	0	0	0	0	0
Samsung Galaxy S6 edge+	0	0	0	0	0	0	0	0	0	0	0	0	0	167	0	0	0	0	0	0	0
Samsung Galaxy S7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	262	0	0	0	0	0	0
Xiaomi Mi A2 Lite	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	276
Xiaomi Redmi 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10703	0	0	0	31	0
Xiaomi Redmi 5 Plus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1887	0	0	0	1664
Xiaomi Redmi Note 7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2567	0	0	0
Xiaomi Redmi Note 9S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	632	0	0

Table 16
Device Classification Confusion Matrix - Datasets: cagliari in mode: PS.

	Apple XS max	Apple iPhone 12	Apple iPhone 6	Apple iPhone 7	Apple iPhone XR	Google Pixel 3A	Huawei CLT-L09 (P20)	Huawei STF-L09 (Honor 9)	Huawei VTR-L09 (P10)	OnePlus Nord	Samsung Galaxy J6	Samsung Galaxy M31	Samsung Galaxy S4	Samsung Galaxy S7	Xiaomi Redmi Note 7	Xiaomi Redmi Note 9S	Generic Template	Moto G5s OR Redmi 5 Plus
Apple XS max	183	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 12	0	644	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 6	0	0	374	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 7	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone XR	0	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 3A	0	0	0	0	0	51	0	0	0	0	0	0	0	0	0	0	0	0
Huawei CLT-L09 (P20)	0	0	0	0	0	0	672	0	0	0	0	0	0	0	0	0	0	0
Huawei STF-L09 (Honor 9)	0	0	0	0	0	0	0	812	0	0	0	0	0	0	0	0	0	0
Huawei VTR-L09 (P10)	0	0	0	0	0	0	0	0	317	0	0	0	0	0	0	0	0	0
OnePlus Nord	0	0	0	0	0	0	0	0	0	256	0	0	0	0	0	0	0	0
Samsung Galaxy J6	0	0	0	0	0	0	0	0	0	0	112	0	0	0	0	0	0	0
Samsung Galaxy M31	0	0	0	0	0	0	0	0	0	0	0	130	0	0	0	0	0	0
Samsung Galaxy S4	0	0	0	0	0	0	0	0	0	0	0	0	3646	0	0	0	0	0
Samsung Galaxy S7	0	0	0	0	0	0	0	0	0	0	0	0	0	143	0	0	0	0
Xiaomi Mi A2 Lite	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	290	0
Xiaomi Redmi 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	238	0
Xiaomi Redmi 5 Plus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	122	307
Xiaomi Redmi Note 7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	245	0	0	0
Xiaomi Redmi Note 9S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0

Table 17
Device Classification Confusion Matrix - Datasets: cagliari in mode: S.

	Apple XS max	Apple iPhone 12	Apple iPhone 6	Apple iPhone 7	Apple iPhone XR	Google Pixel 3A	Huawei ALE-L21	Huawei CLT-L09 (P20)	Huawei STF-L09 (Honor 9)	Huawei VTR-L09 (P10)	OnePlus Nord	Samsung Galaxy J6	Samsung Galaxy M31	Samsung Galaxy S4	Samsung Galaxy S6 edge+	Samsung Galaxy S7	Xiaomi Redmi Note 7	Xiaomi Redmi Note 9S	Generic Template	Moto G5s OR Redmi 5 Plus
Apple XS max	244	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 12	0	1306	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 6	0	0	251	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 7	0	0	0	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone XR	0	0	0	0	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 3A	0	0	0	0	0	139	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei ALE-L21	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei CLT-L09 (P20)	0	0	0	0	0	0	0	1194	0	0	0	0	0	0	0	0	0	0	0	0
Huawei STF-L09 (Honor 9)	0	0	0	0	0	0	0	0	707	0	0	0	0	0	0	0	0	0	0	0
Huawei VTR-L09 (P10)	0	0	0	0	0	0	0	0	0	298	0	0	0	0	0	0	0	0	0	0
OnePlus Nord	0	0	0	0	0	0	0	0	0	0	166	0	0	0	0	0	0	0	0	0
Samsung Galaxy J6	0	0	0	0	0	0	0	0	0	0	0	135	0	0	0	0	0	0	0	0
Samsung Galaxy M31	0	0	0	0	0	0	0	0	0	0	0	0	103	0	0	0	0	0	0	0
Samsung Galaxy S4	0	0	0	0	0	0	0	0	0	0	0	0	0	3486	0	0	0	0	0	0
Samsung Galaxy S6 edge+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	93	0	0	0	0	0
Samsung Galaxy S7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	71	0	0	0	0
Xiaomi Mi A2 Lite	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	294	0
Xiaomi Redmi 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	299	0
Xiaomi Redmi 5 Plus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	112	228
Xiaomi Redmi Note 7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	195	0	0	0
Xiaomi Redmi Note 9S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	71	0	0

Table 18
Device Classification Confusion Matrix - Datasets: bonn & cagliari in mode: AR.

	Apple XS max	Apple iPhone 12	Apple iPhone 13	Apple iPhone 16	Apple iPhone 7	Apple iPhone XR	Google Pixel 3A	Google Pixel 7a	Google Pixel 9	Huawei ALE-L21	Huawei CLT-L09 (P20)	Huawei STF-L09 (Honor 9)	Huawei VTR-L09 (P10)	Motorola Moto G	Motorola Moto G23	OnePlus 12 5G	OnePlus Nord	Samsung Galaxy A15	Samsung Galaxy J6	Samsung Galaxy M31	Samsung Galaxy S24	Samsung Galaxy S4	Samsung Galaxy S6 edge+	Samsung Galaxy S7	Xiaomi Pocophone F1	Xiaomi Redmi 4	Xiaomi Redmi 5 Plus	Xiaomi Redmi Note 7	Xiaomi Redmi Note 9S	Generic Template	Xiaomi Mi A2 Lite OR Redmi 5 Plus
Apple XS max	179	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 12	0	59	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 13	0	0	238	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 16	0	0	0	259	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 7	0	0	0	0	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone XR	0	0	0	0	0	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 3A	0	0	0	0	0	0	123	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 7a	0	0	0	0	0	0	0	1241	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 9	0	0	0	0	0	0	0	0	380	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei ALE-L21	0	0	0	0	0	0	0	0	0	1504	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei CLT-L09 (P20)	0	0	0	0	0	0	0	0	0	0	246	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei STF-L09 (Honor 9)	0	0	0	0	0	0	0	0	0	0	0	317	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei VTR-L09 (P10)	0	0	0	0	0	0	0	0	0	0	0	0	119	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G	0	0	0	0	0	0	0	0	0	0	0	0	0	988	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	197	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OnePlus 12 5G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1009	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OnePlus Nord	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	143	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy A15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	354	0	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy J6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	240	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy M31	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	113	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy S24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy S4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3403	0	0	0	0	0	0	0	0	0
Samsung Galaxy S6 edge+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	167	0	0	0	0	0	0	0	0
Samsung Galaxy S7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	262	0	0	0	0	0	0	0
Xiaomi Mi A2 Lite	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	276
Xiaomi Pocophone F1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1294	0	0	0	0	0	0
Xiaomi Redmi 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10703	0	0	0	31	0
Xiaomi Redmi 5 Plus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1887	0	0	0	1664
Xiaomi Redmi Note 7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2567	0	0	0
Xiaomi Redmi Note 9S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	632	0	0

Table 19
Device Classification Confusion Matrix - Datasets: bonn & cagliari in mode: PS.

	Apple XS max	Apple iPhone 11	Apple iPhone 12	Apple iPhone 16	Apple iPhone 6	Apple iPhone 7	Apple iPhone XR	Google Pixel 3A	Google Pixel 7a	Google Pixel 9	Huawei CLT-L09 (P20)	Huawei STF-L09 (Honor 9)	Huawei VTR-L09 (P10)	Motorola Moto G	Motorola Moto G23	OnePlus 12 5G	OnePlus Nord	Samsung Galaxy A51	Samsung Galaxy J6	Samsung Galaxy M31	Samsung Galaxy S24	Samsung Galaxy S4	Samsung Galaxy S7	Xiaomi Pocophone F1	Xiaomi Redmi 5 Plus	Xiaomi Redmi Note 7	Xiaomi Redmi Note 9S	Generic Template	Motorola Moto G5s OR Xiaomi Redmi 5 Plus	Xiaomi Mi A2 Lite OR Redmi 5 Plus
Apple XS max	183	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 11	0	309	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 12	0	0	644	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 16	0	0	0	285	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 6	0	0	0	0	374	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 7	0	0	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone XR	0	0	0	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 3A	0	0	0	0	0	0	0	51	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 7a	0	0	0	0	0	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 9	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei CLT-L09 (P20)	0	0	0	0	0	0	0	0	0	0	672	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei STF-L09 (Honor 9)	0	0	0	0	0	0	0	0	0	0	0	812	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei VTR-L09 (P10)	0	0	0	0	0	0	0	0	0	0	0	0	317	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	93	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G5s	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	129	0	33
OnePlus 12 5G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	180	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OnePlus Nord	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	256	0	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy A51	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	144	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy J6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	112	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy M31	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	130	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy S24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	78	0	0	0	0	0	0	0	0	0
Samsung Galaxy S4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3646	0	0	0	0	0	0	0	0
Samsung Galaxy S7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	143	0	0	0	0	0	0	0
Xiaomi Mi A2 Lite	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	290	0	0
Xiaomi Pocophone F1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1001	0	0	0	0	0	0
Xiaomi Redmi 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	238	0	0
Xiaomi Redmi 5 Plus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	157	0	0	122	16	130
Xiaomi Redmi Note 7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	245	0	0	0	0
Xiaomi Redmi Note 9S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0	0

Table 20
Device Classification Confusion Matrix - Datasets: bonn & cagliari in mode: S.

	Apple XS max	Apple iPhone 11	Apple iPhone 12	Apple iPhone 13	Apple iPhone 16	Apple iPhone 6	Apple iPhone 7	Apple iPhone XR	Google Pixel 3A	Google Pixel 7a	Google Pixel 9	Huawei ALE-L21	Huawei CLT-L09 (P20)	Huawei STF-L09 (Honor 9)	Huawei VTR-L09 (P10)	Motorola Moto G	Motorola Moto G23	Motorola Moto G5s	OnePlus 12 5G	OnePlus Nord	Samsung Galaxy A15	Samsung Galaxy A51	Samsung Galaxy J6	Samsung Galaxy M31	Samsung Galaxy S24	Samsung Galaxy S4	Samsung Galaxy S6 edge+	Samsung Galaxy S7	Xiaomi Pocophone F1	Xiaomi Redmi 5 Plus	Xiaomi Redmi Note 7	Xiaomi Redmi Note 9S	Generic Template	Motorola Moto G5s OR Xiaomi Redmi 5 Plus	Xiaomi Mi A2 Lite OR Redmi 5 Plus
Apple XS max	244	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 11	0	400	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 12	0	0	1306	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 13	0	0	0	171	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 16	0	0	0	0	376	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 6	0	0	0	0	0	251	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone 7	0	0	0	0	0	0	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Apple iPhone XR	0	0	0	0	0	0	0	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 3A	0	0	0	0	0	0	0	0	139	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 7a	0	0	0	0	0	0	0	0	0	42	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Google Pixel 9	0	0	0	0	0	0	0	0	0	0	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei ALE-L21	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei CLT-L09 (P20)	0	0	0	0	0	0	0	0	0	0	0	0	1194	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei STF-L09 (Honor 9)	0	0	0	0	0	0	0	0	0	0	0	0	0	707	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Huawei VTR-L09 (P10)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	298	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	104	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Motorola Moto G5s	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	10	0	0	155	5	28
OnePlus 12 5G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	183	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OnePlus Nord	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	166	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy A15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	452	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy A51	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	274	0	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy J6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	135	0	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy M31	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	103	0	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy S24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	49	0	0	0	0	0	0	0	0	0	0
Samsung Galaxy S4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3486	0	0	0	0	0	0	0	0	0
Samsung Galaxy S6 edge+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	93	0	0	0	0	0	0	0	0
Samsung Galaxy S7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	71	0	0	0	0	0	0	0
Xiaomi Mi A2 Lite	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	294	0	0
Xiaomi Pocophone F1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1009	0	0	0	0	0	0
Xiaomi Redmi 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	299	0	0
Xiaomi Redmi 5 Plus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	126	0	0	112	0	102
Xiaomi Redmi Note 7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	195	0	0	0	0
Xiaomi Redmi Note 9S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	71	0	0	0

Algorithm 1 Automatically separate devices based on their templates, fingerprints and a supplied knowledge base

Require: Input: P: set of all probe requests; KB: knowledge base with template rules
Ensure: Output: B: final device binning

```
T ← {} // Template bins
for each probe p ∈ P do
    tp ← template(p)
    Append p at T[tp]
end for

for each bin tp ∈ T do
    bin_name ← 'empty' // just to check if bin is known in KB
    devices ← T[tp]
    if |devices| = 1 then
        // Single device case
        tp_rules ← KB[tp]
        for rule in tp_rules do
            Calculate fp from device using rule
            if fp in KB[tp] then
                bin_name ← KB[tp][fp]
                Insert bin as bin_name into B
            end if
        end for

    else
        sep ← findMinimalSeparator(tp, devices)
        if sep = ∅ then
            // Multiple devices - no separator / default bin name
            if tp in KB then
                bin_name = KB[tp]
                Insert bin as bin_name into B
            end if
        else
            // Multiple devices - separator is available
            for dev in devices do
                tp_rules ← KB[tp]
                for rule in tp_rules do
                    Calculate fp from dev using rule
                    if fp in KB[tp] then
                        bin_name ← KB[tp][fp]
                        Insert bin as bin_name into B
                    end if
                end for
            end for
        end if
    end if

    if bin_name is 'empty' then
        Insert bin as 'unknown' into B
    end if
end for

return B
```

References

[1] E. Fenske, D. Brown, J. Martin, T. Mayberry, P. Ryan, E.C. Rye, Three years later: A study of MAC address randomization in mobile devices and when it succeeds., Proc. Priv. Enhancing Technol. 2021 (3) (2021) 164–181.

[2] M. Uras, E. Ferrara, R. Cossu, A. Liotta, L. Atzori, MAC address de-randomization for WiFi device counting: Combining temporal-and content-based fingerprints, Comput. Netw. 218 (2022) 109393.

[3] M. Vanhoef, C. Matte, M. Cunche, L.S. Cardoso, F. Piessens, Why MAC address randomization is not enough: An analysis of Wi-Fi network discovery mechanisms, in: Proceedings of the 11th ACM on Asia Conference on Computer and Communications Security, 2016, pp. 413–424.

[4] L. Yu, B. Luo, J. Ma, Z. Zhou, Q. Liu, You are what you broadcast: Identification of mobile and {IoT} devices from (public){WiFi}, in: 29th USENIX Security Symposium (USENIX Security 20), 2020, pp. 55–72.

[5] I.C.S.L.S. Committee, IEEE standard for information technology-telecommunications and information exchange between systems-local and metropolitan area networks-specific requirements part 11: Wireless LAN medium access control (MAC) and physical layer (PHY) specifications, IEEE Std 802.11^ (2007).

[6] D. Eastlake 3rd, J. Abley, IANA Considerations and IETF Protocol and Documentation Usage for IEEE 802 Parameters, Technical Report, IANA, 2013.

[7] T. Bravenec, J. Torres-Sospedra, M. Gould, T. Fryza, UJI probes revisited: Deeper dive into the dataset of wi-fi probe requests, IEEE J. Indoor Seamless Position. Navig. (2023).

[8] L. Pintor, L. Atzori, Analysis of Wi-Fi probe requests towards information element fingerprinting, in: GLOBECOM 2022-2022 IEEE Global Communications Conference, IEEE, 2022, pp. 3857–3862.

[9] L. Hao, B. Huang, B. Jia, G. Mao, On the fine-grained crowd analysis via passive WiFi sensing, IEEE Trans. Mob. Comput. (2023).

[10] G. Delzanno, L. Caputo, D. D'Agostino, D. Grosso, A.H. Mustajab, L. Bixio, M. Rulli, Automatic passenger counting on the edge via unsupervised clustering, Sensors 23 (11) (2023) 5210.

[11] A.K. Mishra, A.C. Viana, N. Achir, Bleach: From WiFi probe-request signatures to MAC association, Ad Hoc Networks 164 (2024) 103623.

[12] T. Bravenec, J. Torres-Sospedra, M. Gould, T. Fryza, UJI probes: Dataset of wi-fi probe requests, in: 2023 13th International Conference on Indoor Positioning and Indoor Navigation, IPIN, IEEE, 2023, pp. 1–6.

[13] L. Pintor, L. Atzori, A dataset of labelled device Wi-Fi probe requests for MAC address de-randomization, Comput. Netw. 205 (2022) 108783.

[14] M.V. Barbera, A. Epasto, A. Mei, S. Kosta, V.C. Perta, J. Stefa, CRAWDAD dataset sapienza/probe-requests (v. 2013-09-10), Sep, 2013.

[15] D. Vogel, F. Viola, N.M. Kreimeyer, M. Meier, I know you were here: Leveraging probe request templates for identifying Wi-Fi devices, in: 20th International Conference on Wireless and Mobile Computing, Networking and Communications, WiMob 2024, Paris, France, October 21-23, 2024, IEEE, 2024, pp. 429–436, http://dx.doi.org/10.1109/WIMOB61911.2024.10770475.

[16] IEEE Standards Association, IEEE 802.11ac-2013 - IEEE Standard for Information technology–Telecommunications and information exchange between systems–Local and metropolitan area networks–Specific requirements–Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications–Amendment 4: Enhancements for Very High Throughput for Operation in Bands Below 6 GHz, Technical Report, Institute of Electrical and Electronics Engineers (IEEE), 2013, URL https://standards.ieee.org/ieee/802.11ac/4473/.

[17] M. Hirth, T. Hoßfeld, M. Mellia, C. Schwartz, F. Lehrieder, Crowdsourced network measurements: Benefits and best practices, Comput. Netw. 90 (2015) 85–98, http://dx.doi.org/10.1016/j.comnet.2015.07.003, URL https://www.sciencedirect.com/science/article/pii/S1389128615002236 Crowdsourcing.

[18] Google LLC, "Dumpsys | android studio | android developers", 2025, https://developer.android.com/tools/dumpsys, (Accessed: 10 Jan 2025).

[19] E. Lohan, J. Torres-Sospedra, H. Leppäkoski, P. Richter, Z. Peng, J. Huerta, Wi-Fi crowdsourced fingerprinting dataset for indoor positioning, Data 2 (4) (2017-10-03) 32, http://dx.doi.org/10.3390/data2040032, URL https://www.mdpi.com/2306-5729/2/4/32.

[20] Arkasha, Bobzilla, Uhtu, Thuddwhir, Wos, WiGLE.net, 2025, https://wigle.net/faq, (Accessed: 14 May 2025).

[21] Y. Li, J. Barthelemy, S. Sun, P. Perez, B. Moran, A case study of WiFi sniffing performance evaluation, IEEE Access 8 (2020) 129224–129235.

[22] D. Benjamin, Applying Generate Random Extensions And Sustain Extensibility (GREASE) to TLS Extensibility, in: Request for Comments, (8701) RFC Editor, 2020, http://dx.doi.org/10.17487/RFC8701, URL https://www.rfc-editor.org/info/rfc8701 RFC 8701.

[23] R. Rusca, F. Sansoldo, C. Casetti, P. Giaccone, What WiFi probe requests can tell you, in: 2023 IEEE 20th Consumer Communications & Networking Conference, CCNC, IEEE, 2023, pp. 1086–1091.

[24] X. Gu, W. Wu, X. Gu, Z. Ling, M. Yang, A. Song, Probe request based device identification attack and defense, Sensors 20 (16) (2020) 4620.

[25] T. He, J. Tan, S.-H.G. Chan, Self-supervised association of wi-fi probe requests under MAC address randomization, IEEE Trans. Mob. Comput. 22 (12) (2022) 7044–7056.

[26] M.V. Barbera, A. Epasto, A. Mei, V.C. Perta, J. Stefa, Signals from the crowd: uncovering social relationships through smartphone probes, in: Proceedings of the 2013 Conference on Internet Measurement Conference, 2013, pp. 265–276.